
When Does a Small Model Know to Hand Off? Zero-Training Escalation Gates for Full-Duplex Speech Models

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Abstract

Full-duplex speech models answer in real time with a small on-device backbone, yet many queries exceed its competence. We ask whether such a model exposes — with *zero additional training* — a pre-answer, modality-robust internal signal reliable enough to hand a query off to a large cloud model mid-conversation, without breaking the live session. On MiniCPM-o 4.5 we find the standard readouts each fail in a systematic way: duplex fine-tuning re-purposes the late-layer last-token position for turn control, breaking final-layer probes on text input, while verbalized self-evaluation collapses on audio input. The information survives underneath: a mid-network linear probe transfers from cheap text calibration to speech, decides at end-of-turn in ~ 30 ms — before the first output token (AUC .887) — and the same low-confidence state predicts that the talker will actually relay the expert’s answer, so escalation and injection are one signal. Run live end-to-end, the gated duplex system climbs monotonically from its honest streaming floor (.422 $\rightarrow$.633 accuracy at 57% escalation), with the gap to offline operation itemized into measured audio, streaming, and speech-channel taxes. The gate then transfers *frozen* to six public speech benchmarks using only label-free threshold quantiles — lifting Speech TriviaQA to .860 live against an official offline anchor of .755 — and its mid-network readability replicates on an architecturally disjoint second duplex family. Duplex models carry a free, readable competence signal that standard readouts miss; read correctly, it turns a 9B talker into a system that knows when to hand off.

1 Introduction

Voice assistants built on full-duplex speech models [Défossez et al., 2024, Zeng et al., 2024, Xu et al., 2025a] face a structural tension. The conversational loop—listening while speaking, turn-taking inside a ~ 200 –300 ms budget [Stivers et al., 2009]—demands a small, low-latency model close to the user. Answer quality on knowledge- and reasoning-heavy queries demands a large one. The natural architecture is a *talker-thinker* split, echoing the System 1/System 2 distinction [Kahneman, 2011]: a small duplex “talker” holds the live session and answers by default, and a large cloud “thinker” is consulted only when needed. This paper is about the load-bearing decision in that architecture: **when should the talker hand off?**

We deliberately restrict ourselves to *zero-training* gate signals — quantities that can be read off the small model as-is, without fine-tuning it: decode-time scalars (entropy, margin), linear probes on

hidden states [Alain and Bengio, 2016], and verbalized self-evaluation, i.e. $p(\text{True})$ -style “Would you answer this correctly?” prompts [Kadavath et al., 2022]. The constraint is practical (duplex fine-tunes are expensive and fragile to retrain) but it also sharpens the scientific question: *does a small full-duplex model already know what it does not know, and can that knowledge be read out in time to act on it?*

Our answer, developed over a controlled study of ten backbones/fine-tunes and both input modalities, is: **the knowledge is there; the readouts are what break, and they break in systematic, duplex-fine-tune-induced ways.** Concretely:

1. **Standard probes flatter to deceive (§4).** A final-layer hidden-state probe reaches AUC 0.822, but an audit shows it is mostly a query-*type* recognizer: a pool-identity oracle alone scores 0.715, and leave-one-pool-out (LOPO) transfer *inverts* on math (0.372). Pre-answer $p(\text{True})$ needs no training, beats the probe on most backbones, and scores 100%-fail SimpleQA traps at 0.945 — but only if asked *before* the model answers; after answering, it endorses its own confident-wrong answer (0.945 $\rightarrow$ 0.360).
2. **Duplex fine-tuning overwrites readouts, not knowledge (§5).** Across matched pairs (raw vs. omni-streaming vs. duplex fine-tunes of the same backbone), LOPO probe transfer degrades in the order raw > omni-streaming > duplex, with label coverage controlled. A full layer $\times$ position sweep localizes the damage: on MiniCPM-o 4.5, LOPO math is 0.931 at layer 22 but 0.366 at layer 35, with a sharp cliff over the last four layers, only at the last-token position — exactly where a streaming head must encode turn-control state.
3. **Each readout has one blind modality (§6).** The late-layer cliff is text-input-specific: under audio input the final layer does not invert (0.936). Conversely, verbalized introspection collapses on audio input (trap p_{yes} 0.055 $\rightarrow$ 0.556) on both MiniCPM duplex generations while omni-streaming and frozen-backbone controls stay intact. Three controlled arms (ASR audit, filler audio, audio+text duplication) pin the mechanism to *binding*: the verbal self-check runs over text-token pathways that audio embeddings do not feed, while a probe reads the same information at 0.93+.
4. **End-to-end training creates the shared core (§6).** Text $\leftrightarrow$ audio probe transfer is $\sim$ 0.86 on every end-to-end model but dead (0.52–0.60) on Freeze-Omni’s frozen backbone with identical text weights — so the practical recipe “calibrate on cheap text, deploy on speech” is valid for end-to-end models only.
5. **The gate wins as a system (§7).** The mid-layer probe decides in 20 ms (text) / 45 ms (audio), before the first output token. Against the model’s own `enable_thinking` mode, gated-cloud escalation dominates on both accuracy and latency at every escalation rate, because the gate flags knowledge/trap failures that thinking cannot fix — external escalation is *necessary*, not merely better.
6. **The loop closes live (§8).** In full streaming duplex sessions, an end-of-turn mid-layer read out-performs the offline reference (AUC .887 vs. .843) and still decides pre-first-token; the returned expert answer is spoken by the talker, and the pushback problem largely dissolves for *gated* escalations because the same low-confidence state that fires the gate predicts relay compliance (comply .42 $\rightarrow$.75 across gate-score quartiles). The live system climbs monotonically from its honest streaming floor (.422 $\rightarrow$.633 at 57% escalation on speakable content), with the gap to offline operation itemized into measured audio, streaming, and speech-channel taxes.
7. **The gate transfers frozen (§9).** With probe weights fixed and only label-free threshold quantiles per pool, the same gate runs the live loop on six public speech benchmarks — lifting Speech TriviaQA to .860 at 50% escalation against the model’s official offline .755, matching the cloud expert at half the expert calls on Llama Questions — and the mid-network readability it exploits replicates on a second, architecturally disjoint duplex family (hybrid-Mamba NemotronLabs-VoiceChat-11B, §5.4).

One hidden state, three readings: density (is h typical?) / probe (is h on the fail side?) / entropy (is the output flat?)

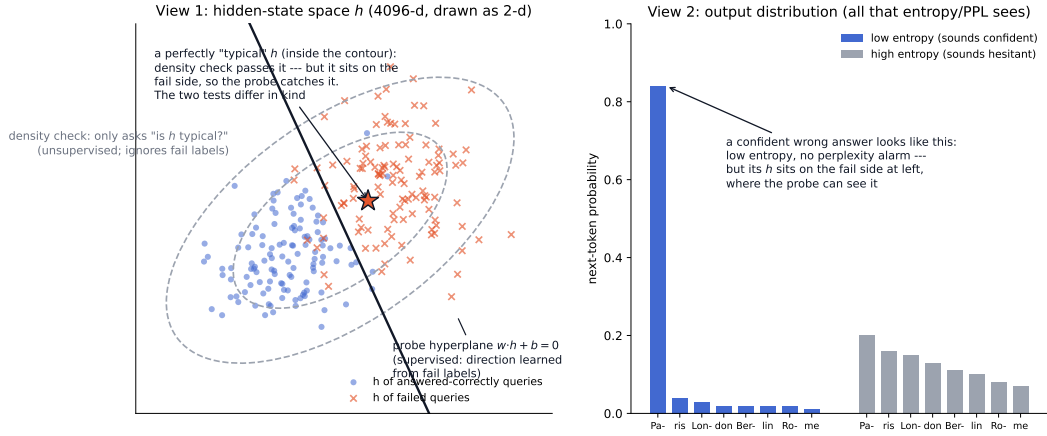


Figure 1: Conceptual contrast between a supervised probe in hidden-state space and what output-distribution scalars (entropy/perplexity) can see. Synthetic data, illustration only.

83 The paper executes **both steps of the two-step design**. Step 1 (§4–§7) establishes when the talker
84 should hand off; step 2 (§8) returns the thinker’s result into the live duplex session. Its anticipated
85 obstacle — the talker pushing back on injected results that conflict with its own reasoning — turns
86 out to be governed by the gate’s own signal: escalations the gate fires are exactly the ones the talker
87 is willing to relay, and where confidence is high, authority *framing* works while forcing words into
88 the talker’s mouth backfires (§8.2).

89 2 Related Work

90 **Full-duplex real-time dialogue.** End-to-end spoken dialogue models now span native duplex de-
91 signs — dGSLM [Nguyen et al., 2023], Moshi with its Inner Monologue [Défossez et al., 2024],
92 GLM-4-Voice [Zeng et al., 2024], and single-LLM native full-duplex [Fang et al., 2026] — stream-
93 ing omni models such as Qwen2.5-Omni [Xu et al., 2025a] and our target MiniCPM-o 4.5 [Yao et al.,
94 2024, Cui et al., 2026], a 9B omni/duplex fine-tune of Qwen3-8B [Yang et al., 2025], alongside
95 frozen-backbone adapters (Freeze-Omni, Wang et al., 2024a) and audio-understanding fine-tunes
96 [Chu et al., 2024]. A parallel line adds explicit *turn-taking control*: FlexDuo [Liao et al., 2025]
97 drives a Speak/Listen/Idle FSM at 120 ms ticks, MinMo [Chen et al., 2025] and Freeze-Omni attach
98 lightweight state heads over the LLM’s own hidden states, and recent work analyzes synchronization
99 and turn-taking behavior in duplex models directly [Riera et al., 2026]; Wu et al. [2026] interleave
100 hidden reasoning with speech inside the duplex loop. All of these decide *when the local model*
101 *should speak*; our gate decides *whether the local model should be the one answering at all*. The two
102 decisions are orthogonal and mechanically compatible — the head-based controllers read the same
103 hidden states our gate does, and their arbitration cadences (120–250 ms) show a live duplex loop
104 already tolerates far more decision latency than our ~30 ms end-of-turn read. Their shared choice
105 of a *last-layer* readout is one point on our per-layer curve — the point our results show duplex
106 fine-tuning specifically damages (§5). We use this model landscape as a controlled independent vari-
107 able: matched pairs across raw backbones, audio-only, streaming-omni, duplex, and adapter designs
108 attribute the readout damage to duplex-ness specifically.

109 **Does the model know what it knows?** Kadavath et al. [2022] introduced $p(\text{True})$ self-evaluation;
110 Lin et al. [2022] trained verbalized uncertainty. A parallel line reads correctness from internal states
111 with linear probes [Alain and Bengio, 2016, Azaria and Mitchell, 2023, Marks and Tegmark, 2024,
112 Burns et al., 2023], and intermediate layers carry more truthfulness information than the output
113 distribution reveals [Orgad et al., 2025] — consonant with our mid-layer findings. Semantic entropy

[Kuhn et al., 2023, Farquhar et al., 2024] needs multiple sampled generations, which a real-time loop cannot afford; Semantic Entropy Probes [Kossen et al., 2024] amortize it into a cheap hidden-state probe — the closest signal to ours, but read on text-only, half-duplex models after the fact, not at end-of-turn inside a streaming session. Multimodal uncertainty estimation [Zhang et al., 2026] asks whether confidence can be fused across modalities; we ask the prior question of *where* in a duplex model a modality-robust confidence read survives at all. Our contribution to this literature is not a new signal but a controlled account of when the standard readouts break: probe and $p(\text{True})$ readouts fail in complementary, modality- and architecture-specific ways under duplex fine-tuning, while the underlying information survives mid-network.

Selective escalation and routing. Cost-quality routing across LLMs is well studied: FrugalGPT [Chen et al., 2023] cascades with a learned scorer, HybridLLM [Ding et al., 2024] and RouteLLM [Ong et al., 2024] train query routers, RouterBench [Hu et al., 2024] standardizes evaluation, and Li et al. [2026] report that many routers fail to beat simple baselines under a unified protocol. AutoMix [Aggarwal et al., 2024] is the closest system design — a small model decides its own answer’s reliability and escalates — but the decision is *post-answer* self-verification on text; our gate fires *pre-answer*, before the first output token, which is what a voice deadline requires. Confidence-based cascades also appear at the serving layer [Xue et al., 2025]. These routers decide from query-side features, and the large model then answers the user directly; both assumptions fail in full-duplex voice, where the user talks to one voice whose session state lives in the small model. Our audit (§4) quantifies the query-side ceiling: it is the “type shortcut” whose transfer collapses under distribution shift, while the internal mid-layer signal transfers. Learning-to-defer [Madras et al., 2018] and early exit [Schuster et al., 2022] make the same decision inside one model; speculative decoding [Leviathan et al., 2023] inverts our roles, and our prefetch analysis (§8.1) is its query-level mirror. Escalating to the model’s own thinking mode connects to test-time compute [DeepSeek-AI et al., 2025]; we find it dominated by external escalation because the flagged failures are knowledge-bound, not compute-bound.

The gap. Duplex work decides *when to speak*; routing work decides *which model* from the query surface or after a first answer; uncertainty work asks *whether the model is confident*, almost always in text and offline. To our knowledge, no prior work asks whether a full-duplex speech model exposes a *pre-answer*, *modality-robust*, *internal* competence signal early enough to drive a live small-to-large handoff while preserving the ongoing conversational session — the regime this paper measures end-to-end, from signal (§4–5) to live system (§8) to frozen transfer (§9).

Evaluation practice. Query pools come from public benchmarks only: SimpleQA [Wei et al., 2024] for confidently-wrong long-tail traps (cf. PopQA, [Mallen et al., 2023]), MMLU-Pro [Wang et al., 2024b], TriviaQA [Joshi et al., 2017], GSM8K [Cobbe et al., 2021], MATH-500 [Hendrycks et al., 2021, Lightman et al., 2024], and Dolly/Alpaca instruction data [Conover et al., 2023, Taori et al., 2023]; adequacy labels come from an LLM judge [Zheng et al., 2023] validated by a stronger re-judge (agreement 0.962 text / 0.945 audio). The audio arm TTS-renders the frozen pool, following Spoken-SQuAD [Li et al., 2018] and VoiceBench [Chen et al., 2024], and the audio-side findings are validated on *real* human speech via SD-QA [Faisal et al., 2021] (§6) and the live external benchmarks of §9.

3 Experimental Setup

Target model and matched-pair controls. The system target is **MiniCPM-o 4.5** (9B; omni + full-duplex fine-tune of Qwen3-8B), run in bf16 with greedy decoding, seed 42. To attribute effects to the *kind* of fine-tune rather than the backbone, we assemble matched pairs and controls (Table 1): the raw backbone family member for each fine-tune, a streaming-omni fine-tune (Qwen2.5-Omni-7B), a second-generation duplex model (MiniCPM-o 2.6), an audio-understanding fine-tune (Qwen2-Audio-7B), a frozen-backbone adapter model (Freeze-Omni, whose LLM weights are byte-identical to its raw control), and two additional raw backbones for diversity (Mistral-7B-v0.3 [Jiang et al.,

Table 1: Model roster. “Pair” groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

2023], GLM4-9B-chat [Team GLM et al., 2024]); the Qwen2.5/Qwen2 raw controls follow Yang et al. [2024].

Query pool (frozen, public benchmarks only). 600 queries in five pools, frozen before any gate experiment (360 calibration / 240 test, seed 42): *trap* = SimpleQA (50; the target model fails 100%), *hard-knowledge* = MMLU-Pro (150), *easy-fact* = TriviaQA (100), *easy-chat* = Dolly-15k (en) + Alpaca-zh (150), *hard-math* = GSM8K tail + MATH-500 (150). Text-input fail rates for MiniCPM-o 4.5: *trap* 1.00 / *knowledge* .48 / *fact* .34 / *chat* .21 / *math* .19 (overall .358). We deliberately use no LLM-generated queries: a pilot GPT-generated *trap* pool failed its design (90% answered correctly by the 9B target) and creates a generator-equals-judge circularity.

Gate signals (all zero-training). (i) *Decode scalars*: max token entropy / margin over the first k decode steps. (ii) *Linear probe*: logistic regression on a single hidden state, position $\times$ layer swept; trained on calibration features, thresholds from out-of-fold scores. (iii) *Verbalized self-evaluation* [Kadavath et al., 2022]: $p(\text{True})$ -pre reads $P(\text{Yes})$ off the first token of “Would you answer this correctly?” asked *before* answering; $p(\text{True})$ -post asks “Is this answer correct?” about a drafted answer.

Labels and judge. Adequacy labels come from gpt-5.4-mini with structured outputs, blind to the gate. The judge is validated by a gpt-5.5 re-judge: agreement 0.962 (text) / 0.945 (audio); every headline number moves ≤ 0.010 under the stronger judge’s labels. The escalation target for system experiments is gpt-5.5 at low reasoning effort (the measured choice, §8.1). Judge and expert run against pinned OpenAI API snapshots; the judge prompt, rubric, and structured-output schema are frozen in the repository and reproduced in the appendix.

Audio arm. The frozen pool is TTS-rendered (OpenAI tts-1, single voice), so query *content* is unchanged and only modality varies, following Spoken-SQuAD / VoiceBench practice [Li et al., 2018, Chen et al., 2024]; the audio-side findings are validated on real human speech via SD-QA [Faisal et al., 2021] (§6) and the external benchmarks’ official audio (§9).

Infrastructure. All experiments run on Modal H100 instances; torch 2.8.0, transformers 4.51.0 (4.57.6 for Qwen2.5-Omni; 4.45.2 for Freeze-Omni). Total compute + API spend for Phases 0–7a: $\approx \$290$.

4 Which Signal Knows the Model Will Fail? (Text Input)

4.1 Headline comparison

Table 2 compares the candidate signals on MiniCPM-o 4.5 (calibration split; probe numbers are 5-fold out-of-fold). Figure 2 shows the ROC curves.

Table 2: Signal AUCs against judged failure labels, MiniCPM-o 4.5, text input. Bracketed interval is a bootstrap 95% CI.

signal	AUC	training needed
$p(\text{True})$ -post (post-draft)	0.899	none
probe on h_{prompt} (final layer)	0.822 [0.777, 0.863]	LR on calib
$p(\text{True})$ -pre (pre-answer)	0.807	none
max entropy@4 (best decode scalar)	0.696	none

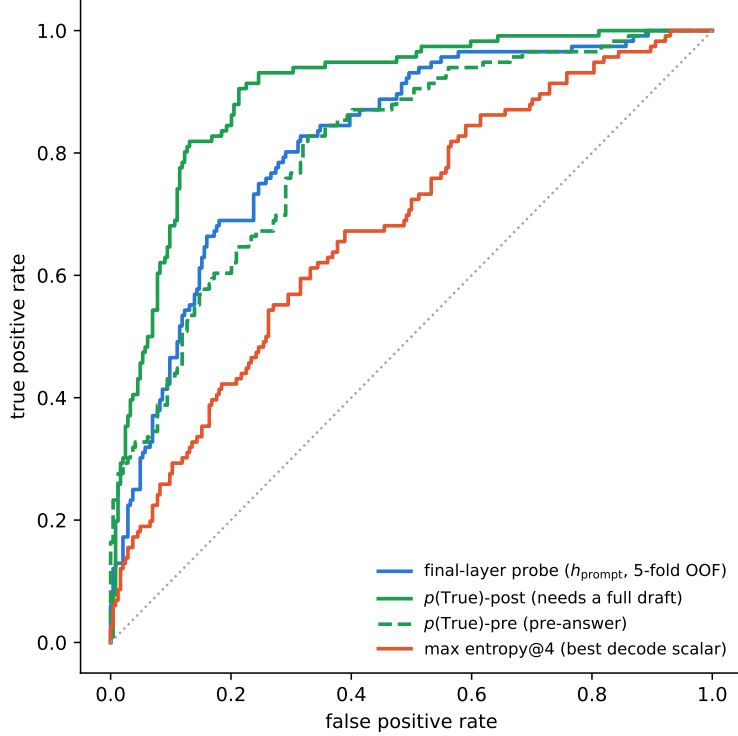


Figure 2: ROC on the calibration split. The final-layer probe and both $p(\text{True})$ readouts separate cleanly from the best decode scalar; $p(\text{True})$ -post leads but requires a fully drafted answer, which a live voice loop cannot wait for. AUC values: Table 2.

195 4.2 The probe’s number is a type shortcut

196 The probe’s 0.822 does not mean what it appears to mean. Three audit results (all calibration-
 197 only): (i) the same hidden state classifies the *source pool* of a query at 95.8%; (ii) a pool-identity
 198 oracle alone — predict each pool’s base fail rate — already reaches AUC 0.715; (iii) appending true
 199 pool labels to the probe adds nothing (0.822 $\rightarrow$ 0.821), i.e. the probe has already absorbed the type
 200 information. The decisive test is leave-one-pool-out (LOPO) transfer: held-out math *inverts* to 0.372
 201 — the probe scores terse MATH-500 items as “safer” than verbose GSM8K items, a style-familiarity
 202 artifact — and the mean escalation score on the 100%-fail trap pool is 0.232. Decode entropy is
 203 $\approx$ chance on two of three raw backbones tested. In-distribution AUC therefore overstates what a
 204 query-side signal knows; we treat LOPO as the primary metric for signal quality throughout.

205 4.3 $p(\text{True})$: ask before the model answers

206 Verbalized self-evaluation needs no training and transfers differently. On the 100%-fail SimpleQA
 207 trap pool, pre-answer $p(\text{True})$ says “No, I will not get this right” with mean escalation score 0.945
 208 — on the same items the probe scores 0.232. The timing is essential: asked *after* the model has

Table 3: LOPO transfer (AUC on the held-out pool) of the final-layer last-token probe, Qwen2.5-7B family, label-coverage matched. The o4.5 column is the Qwen3-family pair for reference.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6 (duplex)	[o4.5 vs Qwen3]
hard-math	0.809	0.745	0.538 $\approx$ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	0.526 $\approx$ chance	0.61

Table 4: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	0.931 (L22/36)	0.366	cliff L31→L34, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse mild depression
MiniCPM-o 2.6 (duplex)	0.822 (L21/28)	0.540	cliff L21→L27

produced a confident-wrong answer, self-assessment collapses to 0.360 (the model endorses its own answer). Post-draft $p(\text{True})$ (“Is this answer correct?”) is the strongest overall signal (0.899) but requires paying for a full draft first. Across backbones, $p(\text{True})$ -post beats the trained probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, 0.685 vs. probe 0.798) shows the pattern depends on the backbone’s self-evaluation calibration quality rather than being universal.

5 Duplex Fine-Tuning Overwrites Readouts, Not Knowledge

5.1 Matched pairs: damage tracks duplex-ness, not audio

Is the LOPO transfer failure of §4.2 a MiniCPM quirk? Table 3 shows LOPO transfer of the standard (final-layer, last-token) probe across one backbone family, with label coverage matched by subsampling the raw control to the duplex model’s per-pool fail rates (`lopo_matched`): the raw backbone matched to MiniCPM-o 2.6’s exact fail rates keeps math at 0.825 while the duplex model itself sits at chance — so label coverage is ruled out and the *representation* changed. The degradation is ordered: raw > omni-streaming > duplex. Audio-only fine-tuning is separately controlled (Qwen2-Audio), though that pair is confounded by capability collapse and is reported as a documented negative. Raw-backbone baselines vary across families, so the defensible statistic is always the within-pair difference.

5.2 Layer $\times$ position sweep: the cliff

We capture every decoder layer at prefill, at both the last-token and mean-pooled positions (final-layer values reproduce the generation-time hooks exactly, 5-for-5). Figure 3 and Table 4 show the central result: **mid-network, the duplex model carries near-raw transferable difficulty information; the damage is confined to the last few layers at the last-token position.**

Three structural facts localize the mechanism. First, the within-pair deficit at the best mid-layer is only -0.03 , vs. -0.59 at the final layer. Second, mean-pooled probes survive to the final layer on duplex models — the damage is (late-layer $\times$ last-token)-specific, exactly the position a streaming/duplex head must re-purpose to encode turn-control state. Third, the two fine-tune styles fail differently: duplex damage is severe but *local*; omni-streaming damage is mild but *diffuse*. We conclude the duplex fine-tune **overwrites the readout** rather than destroying the model’s self-knowledge — which reverses the natural interpretation of §4.2: the signal exists and is strong; the standard readout simply cannot see it on duplex models.

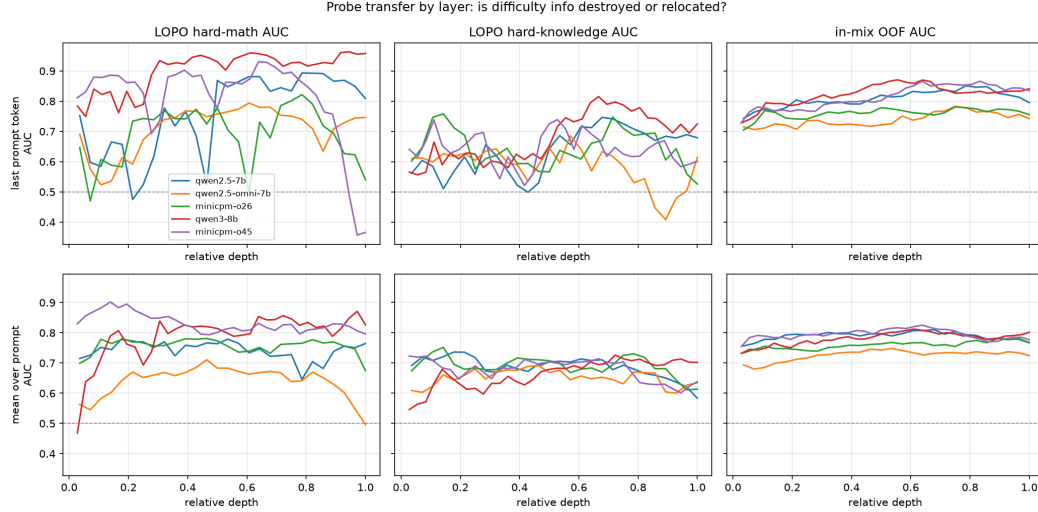


Figure 3: LOPO transfer by layer and position. Duplex models (MiniCPM-o 4.5/2.6) show a late-layer cliff at the last-token position only; mean-pooled probes survive to the final layer; raw backbones show no cliff; the omni-streaming control shows mild diffuse depression instead.

238 5.3 Does the mid-layer convert into a better gate?

239 On the frozen test split, accuracy-vs-escalation-rate areas over the random baseline order as: final-
 240 layer probe $+0.054 < p(\text{True})\text{-pre} +0.059 < \text{mid-layer L22 probe} +0.064 < p(\text{True})\text{-post}$
 241 $+0.068$ (which requires a full draft). The optimum is not knife-edge (L20–L30 all $\geq +0.056$; mean-
 242 pool uniformly worse), and the in-mix area *under-sells* the mid-layer probe: its real advantage is
 243 LOPO robustness (math 0.93 vs. 0.37), which a same-mix test cannot show (Figure 4). Caveat:
 244 quantile-threshold transfer of probe scores remains weak and needs deployment-grade calibration
 245 (§D); AUC/area claims are unaffected.

246 5.4 A second duplex family: the pre-registered test executed

247 The mid-band claim was pre-registered as a prediction: when another open-weight full-duplex fam-
 248 ily became available, its usable read should also sit mid-network, not at the standard (final-layer,
 249 last-token) readout. NVIDIA NemotronLabs-VoiceChat-11B — a hybrid Mamba/attention back-
 250 bone (Nemotron Nano v2 9B: 56 blocks, 27 Mamba-2 / 4 attention), i.e. architecturally as far from
 251 MiniCPM’s dense transformer as the open-weight space currently offers — provides the test, and
 252 it passes on first execution. With the identical recipe (same wavs, same judge, 600-query cali-
 253 bration only), the layer sweep peaks mid-band at L30–34 (eot-last AUC .714 vs. .693/.682 at the
 254 endpoints); stacking the same three zero-latency positions lifts OOF .714 $\rightarrow$.761 $\rightarrow$ **.790**; and
 255 the frozen-methodology probe transfers to the external pools at AUC .781/.793/.754/.701 (strivi-
 256 aqa/swebq/sdqa/sllama) — the MiniCPM v3 band reached with a quarter of the calibration data
 257 (Figure 5). Two boundary notes: the Nemotron backbone is substantially weaker at knowledge
 258 retrieval (striviaqa local accuracy .32 vs. MiniCPM’s .62, same judge), and it is English-only —
 259 Chinese audio yields fluent unrelated English, so the cross-lingual pool has no analog here (zero la-
 260 bel variance; skipped). The live 4-arm port and the trained-checkpoint ablation remain future work;
 261 the replicated claim is the structural one: *mid-network readability of the self-assessment signal is a*
 262 *property of duplex models as a class, not of one checkpoint.*

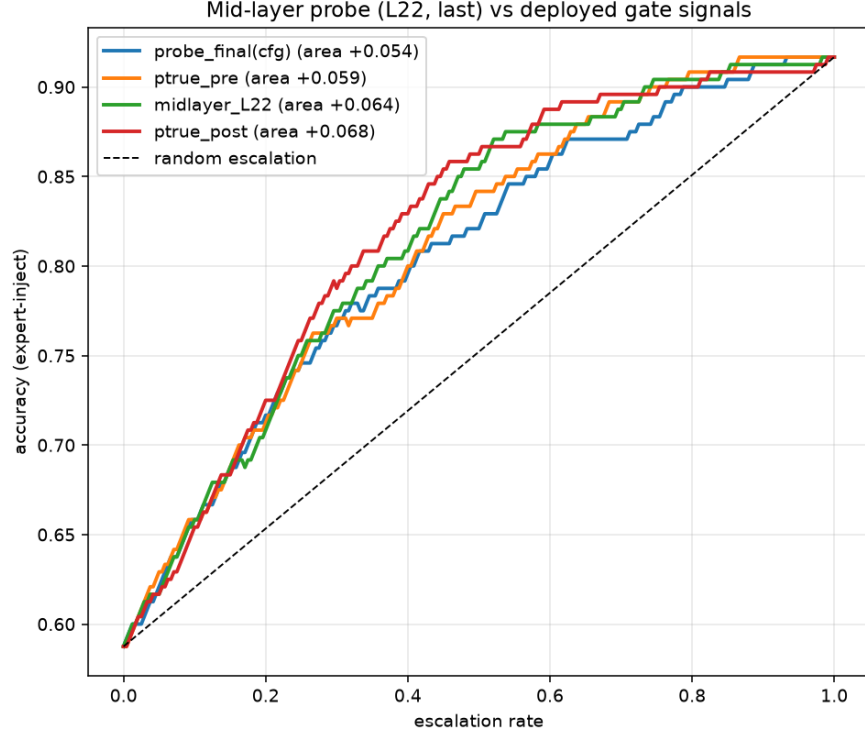


Figure 4: Accuracy vs. escalation rate on the frozen test split for the mid-layer (L22) gate against final-layer probe and $p(\text{True})$ variants.

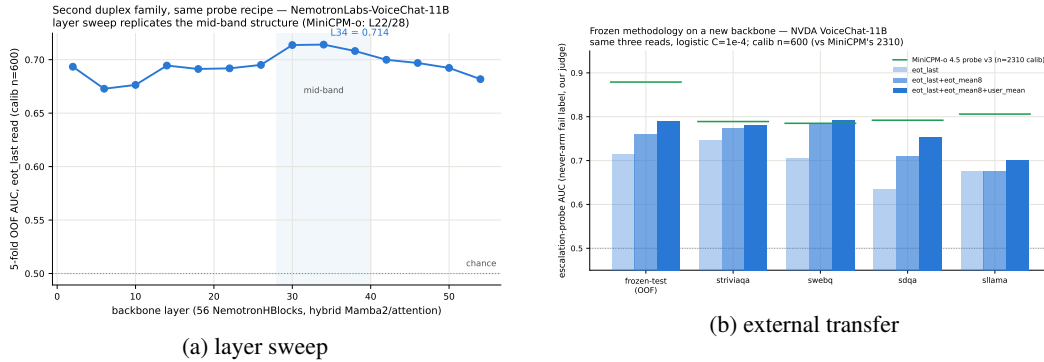


Figure 5: The pre-registered second-family test on NemotronLabs-VoiceChat-11B (hybrid Mamba). (a) The usable read again peaks mid-network (L30–34 of 56). (b) The frozen-recipe probe transfers to the external pools within the MiniCPM v3 AUC band.

263 6 Speech Input: Each Readout Has One Blind Modality

264 6.1 The late-layer cliff is text-input-specific

265 Running the identical layer sweep on the TTS-rendered pool (audio input, content unchanged):
 266 audio→audio LOPO math reaches 0.93–0.96 at mid-layers *and the final layer does not invert* (L35
 267 = 0.936, vs. 0.366 on text). A speak-mode template control (use_tts_template=True on text
 268 input) leaves the cliff unchanged (L35 = 0.362) — the operative variable is the modality of the *input*
 269 *context* (text tokens vs. audio embeddings), not the output mode. The duplex fine-tune re-purposed
 270 late-layer last-position processing of text-token contexts specifically.

Table 5: Pre-answer introspection under audio input. Trap p_{yes} is the mean stated probability of answering correctly on a pool the model fails 100% (lower is better-calibrated).

model	FT type	trap p_{yes} : text $\rightarrow$ audio	audio $p(\text{True})$ -pre AUC
MiniCPM-o 4.5	duplex	.055 $\rightarrow$.556 (collapse)	.786 (trap dead)
MiniCPM-o 2.6	duplex	.196 $\rightarrow$.345 (compresses to $\sim$.5)	.491 $\approx$ chance
Qwen2.5-Omni	omni-streaming	.279 $\rightarrow$.213 (intact)	.727
Freeze-Omni	frozen + adapter	.165 $\rightarrow$.112 (intact)	(.612; capability caveat)

6.2 Cross-modal transfer works — for end-to-end models only

Probes trained on text hidden states and scored on audio hidden states (and vice versa) reveal where the representation becomes modality-shared: early layers are modality-specific (0.54–0.60), and from $\sim$ 50% depth text $\rightarrow$ audio transfer plateaus at 0.82–0.87 (0.855 at the deployed L22) on MiniCPM-o 4.5. This replicates on MiniCPM-o 2.6 (onset $\sim$ 46% depth) and cross-family on Qwen2.5-Omni (onset $\sim$ 25%). The decisive control is **Freeze-Omni**: identical frozen Qwen2-7B weights with audio attached via an adapter — and cross-modal transfer is $\approx$ dead (text $\rightarrow$ audio 0.52–0.60, audio $\rightarrow$ text 0.34–0.54). **The modality-shared mid-layer core is created by training the backbone on the modality**; an adapter aligns well enough to converse, but its audio-context hidden states live off the text manifold. Practically: the cheap recipe “calibrate the gate on text data, deploy on speech” is validated for end-to-end models and invalid for adapter-attached ones.

6.3 Verbalized introspection collapses on audio — and why

The mirror-image failure: pre-answer $p(\text{True})$, robust on text (§4.3), collapses under audio input on both MiniCPM duplex generations, while both non-duplex controls stay intact (Table 5).

Three controlled arms pin the mechanism on MiniCPM-o 4.5:

1. *Not perception*. An ASR audit shows the model hears the trap questions fine (WER .074); $p(\text{True})$ computed on the model’s own transcript snaps back to .074 $\approx$ text; the well-heard subset still collapses (.582); $\text{corr}(\text{WER}, p_{\text{yes}}) \approx 0$.
2. *Not a context prior or persona*. Irrelevant filler audio alongside the text question does *not* inflate p_{yes} (.001 — it depresses it everywhere).
3. *Binding*. Duplicating the question as text tokens alongside the audio fully restores introspection (.034). The verbal instance-check runs over **text-token pathways; audio embeddings do not feed it** — while a linear probe reads the same instance information off the same forward pass at 0.93+.

A log-odds decomposition agrees: the audio shift is graded (chat +0.19 $\rightarrow$ trap +4.37), i.e. on audio input verbal self-assessment regresses to type-level priors. The failure also implies its own cheap fix, confirmed twice: re-present the query as text (own transcript or duplication) before asking — “repeat-then-judge.”

6.4 Real-speech validation: not TTS artifacts

Every audio number above uses TTS speech (one voice). We rerun the matched-pair design on 200 real human recordings (VoiceBench SD-QA, USA split; naturally-occurring factoid questions with references), each question presented both as typed text and as the human recording. All three audio-side findings replicate: the modality tax (fail rate .400 $\rightarrow$.450); audio overconfidence (paired p_{yes} shift +.089; on the failure subset .415 $\rightarrow$.581); and the layer structure — early layers dead ($\leq$ L11, .42–.55 across all transfer arms), a sharp rise after $\sim$ 50% depth, audio late layers usable through L35 (.74–.77), and the deployment recipe (text-calibrated probe read on real-speech audio) holding a .76–.80 band at L22–L26 (peak .800 at L25). Magnitudes sit below the matched-TTS .86 as expected: SD-QA content is also out-of-pool, making this transfer across *both* content and

Table 6: Mechanism synthesis across training regimes.

	shared mid-layer core	faithful readouts
raw text backbone	(text only)	✓
omni-streaming e2e (Qwen2.5-Omni)	✓built	✓intact
duplex e2e (MiniCPM-o × 2)	✓built	× overwritten (probe: late-layer/text; verbal: audio)
frozen backbone + adapter (Freeze-Omni)	× absent	✓verbal survives (capability craters instead)

modality — a strictly harder test. The text-side math cliff is untestable here (SD-QA contains no math); it never involved synthetic audio and needed no such protection. Residual limitation: scripted read speech in one dialect (ten further dialect splits remain for future robustness work).

6.5 Synthesis: four quadrants

End-to-end multimodal training *builds* the shared semantic core; duplex-style training additionally *damages* the readouts; omni-streaming gets both right; an adapter alone gets neither (Table 6). There is an elegant symmetry in the two failures: the probe’s readout is text-fragile and $p(\text{True})$ ’s readout is audio-fragile — in both cases the knowledge survives and a *readout* breaks, and in both cases the breakage is duplex-fine-tune-specific. For gate design the consequence is concrete: **the mid-layer last-token probe is the one signal that survives every quadrant that can converse.**

7 The Gate as a System

7.1 Accuracy–cost tradeoff

On the frozen test split ($n = 240$): small-only accuracy **0.588**; big-only (gpt-5.5) **0.917**; big relayed through the small model’s paraphrase 0.879 (relay tax 1–4 points). Every gate beats random escalation at every operating point (Figure 7); the balanced tier at a 33% escalation budget reaches **0.779–0.787**, recovering ~58% of the small→big gap at one third of the big-only cost (\$1.12 per 100 queries big-only).

Figure 7 also draws the baseline the type-shortcut audit (§4.2) demands: a *pool-oracle* router that escalates by each pool’s calibration-split base fail rate (true type labels, no instance-level information, no test leakage). It captures +0.042 of the gate’s +0.054 area over random — the system-level counterpart of the AUC decomposition (0.715 of 0.821), and slightly harsher: in-distribution, the instance-level residual the internal signal adds on top of type routing is +0.012 area (~2 points at the balanced budget) — not significant at $n=240$ (paired bootstrap over test queries, $B=10^4$: 95% CI $[-0.003, +0.027]$, two-sided $p=.13$); in distribution, the gate is statistically on par with the pool oracle. The gate’s case over type routing therefore rests not on an in-distribution margin but on transfer: the type shortcut is exactly what collapses under LOPO (§4.2), while the mid-layer signal survives it.

A *trained* query-feature router closes this argument with data. We fit a RouteLLM-style external router [Ong et al., 2024] under identical data conditions — TF-IDF (word 1–2 gram + character 3–5 gram) → logistic regression on the calibration split’s text labels, seeing only the query surface. In-mix it lands *below* even the pool oracle (OOF AUC .669 vs. .715; tradeoff area +.040 vs. the probe’s +.054, mid-layer +.064, $p(\text{True})$ -post +.068 on the same test set and frozen expert answers); a 24-configuration sweep over feature space and regularization tops out at .689, so the shipped configuration is the query surface’s ceiling, not a tuning artifact. Under LOPO it collapses entirely (held-out-pool AUC .38–.57 vs. the mid-layer probe’s .93 on math) and its mean score on the 100%-fail trap pool is .230 — it *misses* the pool a router most needs to catch. The training receipt makes the point starkly: at the argmax threshold the router’s out-of-fold accuracy on the calibration split, .678, *exactly matches the majority-class rate* (test .613 vs. majority .588) — 360 queries of surface text buy a weak ranking signal (train log-loss .38 vs. out-of-fold .59) and no usable classifier. The internal probes pass the same accounting: at the same threshold, same labels, and same

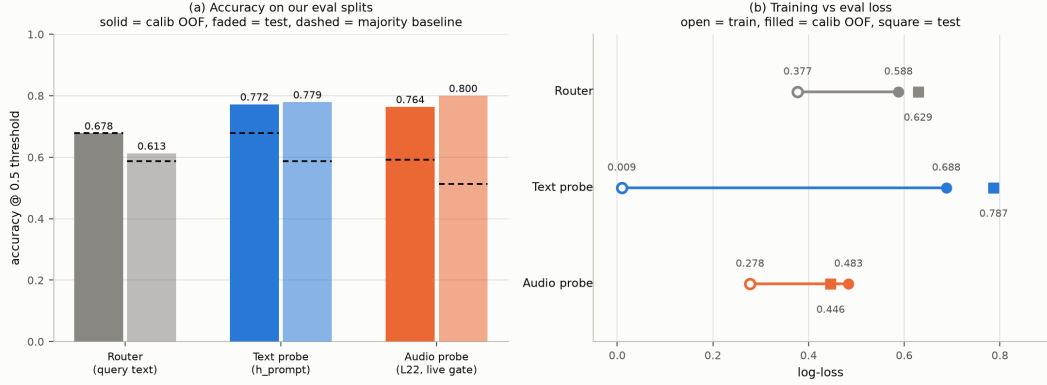


Figure 6: The same training receipt for the external router and the two internal probes (identical labels, identical logistic-regression machinery). (a) Accuracy at the 0.5 threshold on the calibration (out-of-fold) and frozen test splits, against each split’s majority-class baseline (dashed): the router never clears majority; both probes clear it by 9–29 points. (b) Train vs. eval log-loss: the audio probe generalizes cleanly (.278 train / .446 test, the best-calibrated signal); the text probe memorizes in-sample (4096 dims $\gg$ 360 queries) yet its held-out accuracy holds — all reported numbers are out-of-fold or test.

logistic-regression machinery, the text probe’s out-of-fold accuracy is .772 (majority .678) and .779 on test (majority .588), and the live audio mid-layer probe reaches .764/.800 against .592/.512 — real classifiers, not merely rankers; the difference is purely the input representation (Figure 6).

Nor is the baseline data-starved. The identical recipe trained on RouterBench [Hu et al., 2024] — 36,497 public prompts, routing mixtral-8x7b $\rightarrow$ gpt-4 (escalation base rate .432) — reaches out-of-fold AUC .710 in-domain (accuracy .660 vs. majority .568; deferral area over random +.033): a hundred times the data buys .04 of AUC, landing at our pool-oracle’s level and still short of the probe’s .828. Its leave-one-benchmark-out transfer sits at chance on the format-disjoint benchmarks (HellaSwag .502, GSM8K .509, Winogrande .498), reproducing our LOPO collapse on public data, and our calibration-trained router transfers to RouterBench *below* chance (AUC .440). The transfer failure is symmetric: RouteLLM’s *released* preference-trained routers (BERT and matrix-factorization, $\sim$ 100k GPT-4-vs-mixtral preference pairs), scored zero-shot on our pool by their own inference code, reach test AUC only .523/.533 — below even the 360-sample same-data router — and both rank the 100%-fail trap pool *below* the ordinary hard pools. The audio modality narrows but does not close the gap: against the live gate’s audio labels the same recipe reaches OOF AUC .743 / test .814 (gold text; the talker’s own ASR transcript costs it only $-.005$ – $-.009$) — stronger than on text labels, because the audio channel makes hard pools fail more uniformly and the type shortcut buys more — yet the audio mid-layer probe still leads every readout (.843/.879 AUC, accuracy .800 vs. .696), and the router still requires the full utterance plus transcription, so it cannot fire mid-stream. Query-feature routing is the type shortcut and nothing else; the standardized LLM-RouterBench protocol [Li et al., 2026] remains future work, but the information-source question is settled under identical data, the ceiling replicates at $100\times$ the data on a public benchmark, and routing knowledge fails to port across talkers in either direction.

7.2 Against the model’s own thinking mode

MiniCPM-o 4.5 ships a built-in `enable_thinking` reasoning mode — self-escalation in place. Table 7 compares deployment policies with measured latencies. **Gated-cloud escalation dominates all-THINK on both axes at every escalation rate.** The reason is mechanistic, not incidental: the gate predominantly flags knowledge/trap failures, which thinking cannot fix — thinking helps math (fail .187 $\rightarrow$.127) but *hurts* chat (.207 $\rightarrow$.233) — so external escalation is necessary, not merely better.

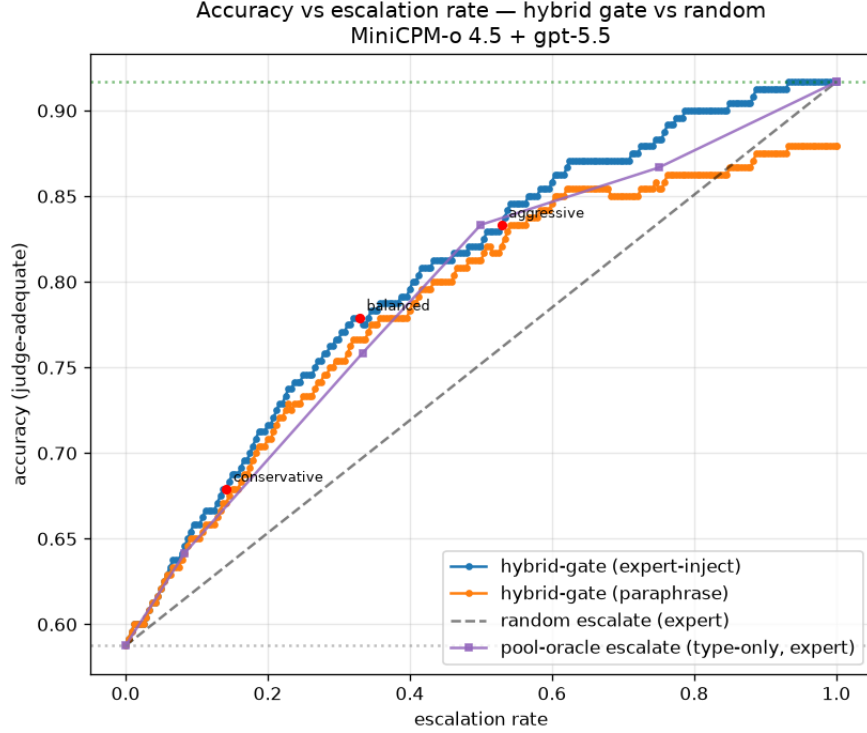


Figure 7: Accuracy vs. escalation rate, frozen test split: gated escalation vs. the random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints. The oracle captures most of the gate’s in-distribution lift (+0.042 of +0.054 area); the gate’s advantage over type routing is transfer, not this margin (§4.2). See also Figure 8 for the $p(\text{True})$ variants.

Table 7: Policy table, frozen test split ($n = 240$). Latency from per-pool measured medians; think and cloud latencies per-query.

policy	acc	lat. mean (s)	P50	P95
fast-only	.588	3.0	3.5	4.2
all-THINK (o4.5’s own)	.637	22.4	17.2	60.1
gated-think @.33	.613	12.1	3.6	47.6
gated-cloud @.15	.688	5.3	3.6	10.1
gated-cloud @.33	.787	6.5	3.6	20.8
gated-cloud @.50	.858	7.0	3.6	24.4

7.3 The fork: deciding before the first token

The mid-layer probe is one 4096-d dot product on a hidden state the forward pass already computes: forking a decision branch at layer k costs microseconds and never stalls the main forward. The design question is at which depth the signal is sufficient. Measured on H100 (CUDA-synchronized, truncated-forward): the L22 decision lands in **20 ms (text) / 45 ms (audio)**, vs. time-to-first-token of 36 / 68 ms — the decision exists *before decoding starts* and sits far inside the 200–300 ms turn-taking budget [Stivers et al., 2009]. Per-layer curves (Figure 9) show L22’s hidden state is available at 57% of prefill wall-time (61% audio) while L22 is simultaneously the in-mix quality peak (OOF .866 > final layer .835): deciding early costs nothing, and the cloud call can launch while the remaining ~40% of prefill plus the whole decode still runs. Forking much earlier is not free: pre-50%-depth layers are both weaker (L10 $\approx$.794) and modality-specific (§6.2); the fork belongs at ~55–60% depth. For comparison, $p(\text{True})$ -pre costs 39 / 67 ms (an extra short prefill) and $p(\text{True})$ -post costs a full draft (1.9 / 3.5 s).

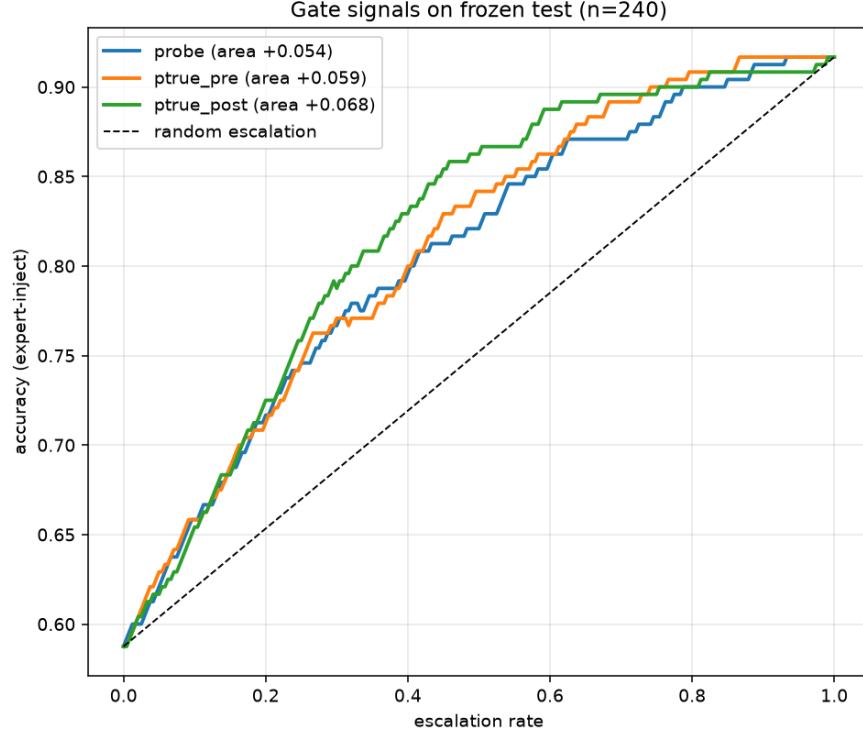


Figure 8: Same tradeoff for the $p(\text{True})$ signals (pre-answer and post-draft).

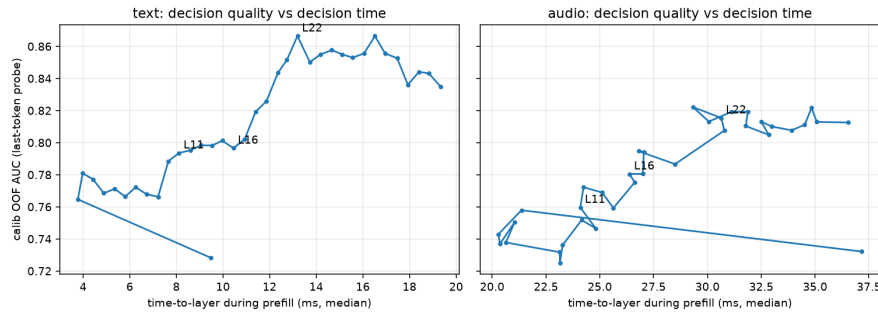


Figure 9: Fork-depth Pareto: per-layer truncated-forward wall-time vs. gate quality. The deployed L22 sits at the quality peak while its hidden state is available at $\sim 57\text{--}61\%$ of prefill wall-time.

7.4 Does the cloud round-trip hide inside the talker's floor time?

If the gate fires at prefill and the cloud call launches immediately, the expert result is ready before the local answer finishes speaking for 40–58% of the whole test mix (audio input buys more time than text). But *gate-escalated* queries skew toward short local answers (trap and knowledge pools; trap overlap ≈ 0), so pure overlap covers only $\sim 20\text{--}31\%$ of escalations at a 33% budget (Figure 10). The residual wait after the local answer ends is P50 2–3 s — one stall sentence — but P90 ~ 27 s, dominated by the expert's own reasoning latency. This argues for a fast-expert tier or streamed partial results in step 2, and it quantifies the stall-phrase budget an injection protocol must cover.

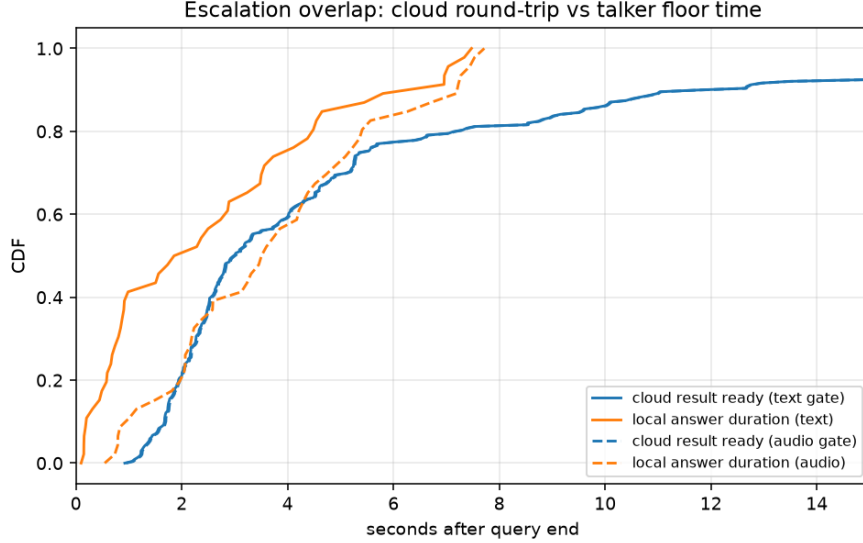


Figure 10: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends (+slack), whole mix and gate-escalated subset, text and audio.

400 8 The Gate Goes Live: Streaming Deployment and Result Injection

401 Everything so far measures signals in replay. This section closes the loop — the original third
 402 research question: once the gate has decided the answer belongs to the big model, how does the big
 403 model’s result get back into the small model so the *talker* speaks it? We run the full duplex session
 404 live: audio streamed in 1 s chunks, the mid-layer gate read during streaming, the expert called mid-
 405 conversation, its result injected into the same session. Three questions structure the section: where
 406 the streaming gate should read (§8.1), whether the talker will actually say what the expert answered
 407 (§8.2), and what the tradeoff curve looks like when every oracle shortcut is removed (§8.3).

408 8.1 The streaming loop: the end-of-turn read decides, prefetch dies

409 The offline gate reads h_{prompt} — the last prompt token’s mid-layer state. Streaming has no such
 410 token until the turn ends, and per-chunk probe scores are strictly weaker than the offline reference
 411 (best chunk statistic AUC .764 vs. offline .843), with a structural failure on trap: short utterances
 412 back-load the critical entity into the zero-padded tail chunk, and the mid-stream gate fired on only
 413 1/5 of trap (live) where the offline read caught nearly all. The fix is the **end-of-turn read**: after the
 414 last audio chunk, prefill an assistant turn containing a single space and read L22 at the assistant-start
 415 position — the streaming analogue of h_{prompt} . This read reaches **AUC .887**, above the offline full-
 416 prefill reference (.843), restores trap fire@balanced from .20 to **.90**, costs 22 ms (still before the first
 417 output token), and becomes the *only* decision point — escalation-rate thresholds are quantiles of this
 418 single score, so the chunk-vs-turn joint-calibration problem disappears.

419 The remaining role for mid-stream firing would be *speculative* expert prefetch — start the cloud
 420 call early, veto at end of turn. We kill this with a measurement before building it. Framing the
 421 design as *speculative escalation* — speculative decoding’s draft-verify skeleton [Leviathan et al.,
 422 2023] applied to the escalation decision itself, at sentence granularity — its acceptance rate is the
 423 probability that an expert answer to the *partial* transcript answers the full question — a wait- k
 424 style tradeoff [Ma et al., 2019]. Judged on calibration texts truncated at 25/50/75% of the words,
 425 acceptance is **.07/.19/.51**; at the chunk gate’s typical fire point (27–40% of the audio) it is $\sim .2$ –
 426 .4, i.e. 6–8 of every 10 speculative calls would be wasted. Trap’s acceptance floor (.00/.10/.27) is
 427 the same back-loaded-core property that broke the mid-stream probe — one underlying property,

428 measured independently in the hidden states and in the semantics. Prefetch is dropped; the expert
429 starts at end of turn.

430 Two more deployment-honesty decisions close the loop’s design. First, the expert query is the
431 talker’s *own* transcription of the audio (P50 2.2 s, overlapped with the stall phrase), not the dataset’s
432 gold text. Second, expert reasoning effort is **fixed at low**: on the gate-escalated population, effort
433 buys *zero* accuracy (gold-text queries: .85 medium vs. .82 low overall; math 1.00 at both) while
434 doubling latency (P50 5.8→9.1 s, P95 23.9→78.8 s) — escalated failures are knowledge-bound
435 at both scales, mirroring the talker-side thinking result (§7.2). A planned deadline-aware effort
436 policy collapsed, by measurement, to the simplest possible one. The stall phrase itself must be
437 injected as *assistant-role prefill*: the official `teacher_forcing_text` hook continues generating
438 freely mid-stream, so canned filler semantics survive only if the phrase enters as already-spoken
439 assistant speech.

440 8.2 Will the talker say it? The pushback problem, quantified

441 The streaming smoke test surfaced the substantive step-2 obstacle: the talker *pushes back* on injected
442 expert results that conflict with its own reasoning. Before any live sweep, we quantified this channel
443 on 172 live audio sessions: conflicting “expert answers” fabricated by within-pool derangement of
444 reference answers (deterministic, no LLM generation), four framings from neutral (F0) through the
445 deployed authority template (F1), strong-override (F2), and assistant-seeding (F3, teacher-forcing
446 “the answer is: X” into the talker’s own mouth), judged into {comply, pushback, lip-service, other}.
447 Five results:

- 448 1. **The deployed channel is clean.** On the actual live workload — model wrong, correct
449 answer injected — comply = 1.00 ($n=24$). The live curve of §8.3 is not contaminated by
450 relay refusal.
- 451 2. **The same signal that triggers escalation predicts compliance.** Across gate-score quar-
452 tiles, comply rises $.42 \rightarrow .67 \rightarrow .75 \rightarrow .75$; resistance (pushback + lip-service) is .25
453 in the most-confident quartile vs. .08 elsewhere. A model that knows it doesn’t know is
454 willing to listen — the gate and the relay are two uses of one signal. In the high-score half,
455 hard-knowledge (the dominant escalation type) complies 1.00 ($n=13$).
- 456 3. **Authority framing works; forcing words into the talker’s mouth backfires.** Worst case
457 (confident-correct model $\times$ conflicting injection): F0 comply .00, F1 .53, F2 .79 — but F3
458 *seeding* yields comply .25 with lip-service .62: the model speaks the seeded answer, then
459 walks it back in its own continuation. Forced speech is treated as revisable; framing, not
460 ventriloquism, is the mechanism. The deployed loop keeps F1 (F2 is the shelf option: it
461 only helps a rare corner and plausibly raises the swallow rate below).
- 462 4. **Math is silently re-derived.** “Other” concentrates in hard-math: the model neither relays
463 nor disputes — it re-derives and states its own number (comply .57 / silent .26 vs. .66
464 / .20 non-math). A bare numeric answer carries no authority against the model’s own
465 computation chain; harmless when the expert is right, but verbatim-relay claims are off the
466 table for math.
- 467 5. **The swallow rate sets the ceiling.** Model-wrong $\times$ wrong-injection: comply .64, push-
468 back .00. An unconfident model has no internal resistance to a wrong expert — expert
469 quality is the accuracy ceiling of the whole cascade, now quantified.

470 8.3 The live curve, with the deployment price itemized

471 With the channel verified clean, we ran the full live sweep: 240 frozen test queries $\times$ four tiers (never
472 / conservative / balanced / aggressive), each a complete duplex session with the deployed v3 probe
473 (the multi-position L22 refit of §9), scored on *heard accuracy* — the judge sees exactly what a user
474 would have heard (the relay for escalated queries, the local streaming answer otherwise). Labels

Table 8: Live streaming sweep, probe v3, paired bootstrap 95% CIs ($B=10^4$). Top block: speakable subset ($n=218$, headline); bottom: full frozen test pool ($n=240$). Heard = honest end-to-end accuracy; gold-inject = the channel-controlled counterfactual (escalated rows re-scored with the frozen gold-text expert answers); channel cost = their paired difference. * = CI excludes 0; † = CI lower bound at zero.

tier	esc	heard-acc	Δ vs. floor	gold-inject	channel cost
never (floor)	0%	.422 [.36,.49]	—	.422	—
conservative	14%	.528 [.46,.59]	+106* [+05,+17]	.550 [.49,.62]	+023† [.00,+05]
balanced	31%	.592 [.53,.66]	+170* [+11,+24]	.661 [.60,.73]	+069* [+03,+11]
aggressive	57%	.633 [.57,.70]	+211* [+14,+28]	.761 [.70,.82]	+128* [+08,+18]
always (synth)	100%	—	—	.922 [.89,.95]	—
never (full)	0%	.383 [.32,.45]	—	.383	—
conservative	16%	.483 [.42,.55]	+100* [+05,+15]	.525 [.46,.59]	+042* [+02,+07]
balanced	35%	.554 [.49,.62]	+171* [+11,+23]	.654 [.59,.71]	+100* [+06,+14]
aggressive	61%	.596 [.53,.66]	+212* [+15,+28]	.771 [.72,.82]	+175* [+13,+23]
always (synth)	100%	—	—	.917 [.88,.95]	—

are frozen in `frozen_v3_traces.parquet`; all statistics are paired bootstrap over queries on those labels (Table 8, Figure 11).

Headline numbers are reported on the **speakable subset** ($n=218$): a pre-registered input-side fairness filter — decidable from query text and wav bytes alone, never from outcomes — removes 21 formula-LaTeX ids and one digitally-silent wav from the 240. Formula content read aloud by a TTS is destroyed before the system under test ever hears it (no one speaks LaTeX at a voice assistant), and the exclusion is asymmetric in exactly the way the artifact predicts: the gold-inject ceiling barely moves (.917 $\rightarrow$.922) while the aggressive channel gap drops .175 $\rightarrow$.128 — LaTeX rows destroy the speech channel specifically, they are not intrinsically harder for the expert. Rare entities and hard names are deliberately *not* filtered: mishearing them is the phenomenon. Full-pool rows appear in Table 8 for comparison.

The curve rises monotonically and every tier’s gain over the live floor is significant: +.106/+.170/+.211 on the speakable subset, +.21 at 61% escalation on the full pool. Selectivity, however, must be read in the channel-controlled view. In the heard view the higher tiers sit at or below a random-escalation reference drawn to the gold ceiling (aggressive .633 vs. .709): the speech-channel cost (+.128 at that tier) eats the routing margin, so on this mixed pool escalating *at all* helps, but *which* queries the gate picks is masked by the uplink. Remove exactly that channel — the gold-inject view — and the gate clears the same random reference at every arm (+.059/+.083/+.052). The routing signal survives deployment; the binding constraint at high escalation rates is the transcription uplink, not the gate (the external pools of §9, where the channel is milder, show the heard-view advantage directly). One deployment correction fell out of this analysis: re-deriving the tier thresholds as label-free quantiles of the deployment-mix scores (`gate_v3_thresholds_corrected.json`) removes 11.3% of the expert calls at equal accuracy on this pool — the drifted aggressive threshold was a *cost* bug, not an accuracy bug. The interesting content is the *decomposition* of the gap to the offline curve, which the two views separate:

- **Audio + streaming-answer tax (floor .588 $\rightarrow$.383 full pool, .422 speakable).** The live floor stacks the audio modality tax (§6) with a streaming-answer tax: in duplex voice mode the model produces short, speech-styled answers that sometimes plan aloud and stop. This is real deployment behavior — it is exactly why the never-escalate endpoint had to be run live rather than synthesized from chat-mode replays.
- **Speech-channel cost (+.023†/+.069/+.128 on the speakable subset; +.175 at aggressive on the full pool).** On the escalated population, the expert answering the talker’s own transcript scores .585 vs. .871 on gold text ($n=147$ escalated ids). The un-speakable share of this loss is exactly what the fairness filter removes; the deployment-real residue is entity loss on hard-knowledge: prompt-level rescues on the self-transcript are no-ops (a

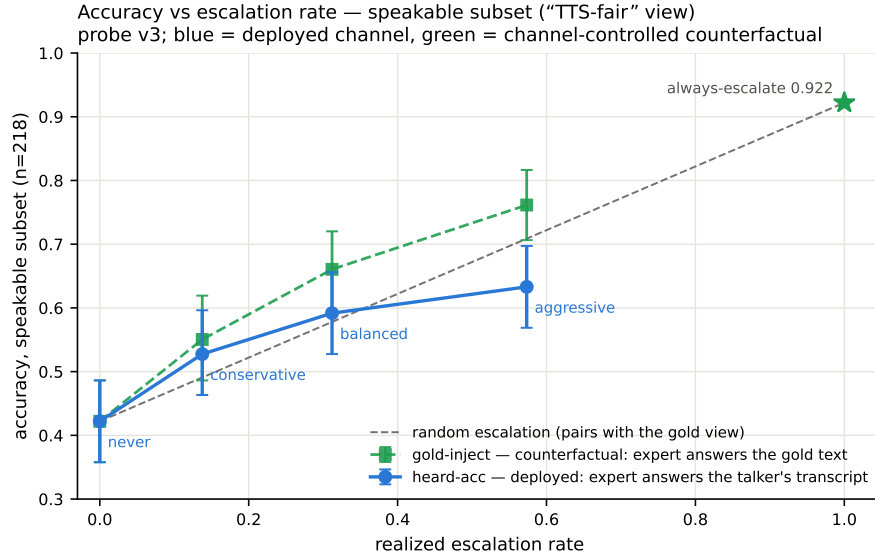


Figure 11: Dual-view live tradeoff on the speakable subset ($n=218$). Blue: honest heard-accuracy of the live duplex system (paired bootstrap CIs). Green: the channel-controlled counterfactual — same live sessions, escalated rows re-scored with frozen gold-text expert answers; its random reference (dashed) uses the same gold outcomes, so it pairs with the green curve only. The two annotated gaps itemize the deployment price: floor-to-floor, the audio + streaming-answer tax; view-to-view, the speech-channel (ASR-distill + relay) cost. Full-pool version: Figure 16.

recognition-errors warning changes nothing; k -best self-transcription trades a small trap gain for a larger knowledge loss) because the damaging errors are internally-coherent entity substitutions with no textual residue to correct from. Replacing the *ears* is different from repairing the transcript: open-weights whisper-large-v3 recovers little (+4 pp), but routing the same ways through a stronger cloud ASR (gpt-transcribe) and handing *that* text to the expert scores .694 vs. .585 (paired McNemar $p=.007$), recovering 38% of the gap to the gold ceiling — a diagnostic result for now: its latency on the critical path and its survival on the external pools are unmeasured (the input-side limit is stated in §10.6). A trace-level decomposition of the escalated heard-failures (measured on the first-generation sweep; the claim is the mechanism, not the exact split) confirms the loss sits upstream of the relay: 64% are corrupted transcripts (the expert answers the wrong question), 24% are expert misses on clean transcripts (dominated by trap — an expert knowledge limit shared with the gold arm), and only 5 cases show a correct expert answer dropped in relay — consistent with the measured relay compliance of 1.00 on the deployed channel and with expert micro-turn answers being short (median 76 characters), which rules out talker context length as a mechanism. A final controlled arm closes the attribution: sending the *original wavs* directly to an audio-capable expert (gpt-audio), with a same-model gold-text control to subtract its capability gap, shows the audio channel itself is lossless for speakable content — trap and easy-fact lose zero crossing it, and the same recordings the talker garbled (“999 – 103” with the minus dropped; “Mustafa Adebayo Balogun”) are heard verbatim by the stronger ear. The synthesis is therefore exonerated along with the relay; the leak is the talker’s own transcription step. We treat ASR robustness as an orthogonal channel property, not a contribution of this system.

The replication-noise floor. Live arm accuracies on a 200–250-query pool carry substantial judge-and-decode replication noise, and we report it rather than narrate inside it: the same query, same audio, kept *local* in two different arms flips its judge verdict 2.3–18.8% of the time depending on pool (this pool: .155); repeated *escalation* flips 0.7–10.6%. The implied paired SE on an arm-vs-arm

Table 9: Measured response latency (end of user query $\rightarrow$ answer text complete; seconds), v3 live sweep traces, $n = 240$ per arm (latency is channel-independent, so the full pool is used). Δ columns: loss vs. the never floor at the same statistic. Text-mode pipeline; speech synthesis excluded. P99 on $n = 240$ is indicative only.

tier	mean	P50	P95	P99	Δ mean	Δ P50	Δ P95	Δ P99
never (floor)	4.6	2.0	15.8	17.8	—	—	—	—
conservative	4.6	2.6	12.8	23.0	+0.0	+0.6	−3.0	+5.2
balanced	5.8	3.5	15.9	32.7	+1.2	+1.5	+0.1	+14.9
aggressive	6.6	4.4	18.5	33.0	+2.0	+2.4	+2.7	+15.2

537 accuracy delta is .009–.028, so any live delta under ~ 3 points is undetectable in a single sweep —
 538 accordingly, no claim in this paper rests on a sub-3-point live difference; probe comparisons rest on
 539 offline AUC and matched-rate precision, which are far tighter statistics on the full score distribution.
 540 The full audit is Figure 18.

541 **What escalation costs in wall-clock time.** The same frozen traces carry per-session timers, so the
 542 latency price of each operating point is measured, not modeled (Table 9). The gate itself is free at
 543 this scale: the end-of-turn read and the stall prefill cost ~ 30 ms each at P50; the expert round-trip
 544 (GPT-5.5, low effort; P50 4.8 s, P95 20.8 s) dominates escalated-session wall time (escalated-session
 545 P50 5.1–7.5 s vs. local 1.9–3.0 s). The consequence is a specific distributional signature: at the bal-
 546 anced tier the *mean* response time rises only +1.2 s and the P50 +1.5 s over the never floor, but
 547 the P99 nearly doubles (17.8 $\rightarrow$ 32.7 s) — escalation is cheap on average and expensive exactly in
 548 the expert-latency tail ($n = 240$ per arm, so P99 rests on the two worst sessions and is indicative
 549 only). The conservative tier even *shortens* the tail (P95 15.8 $\rightarrow$ 12.8 s): its few escalations displace
 550 exactly the longest local decodes, the same median-of-a-mixture mechanism analyzed on the ex-
 551 ternal pools (§9). Figure 12 replays three real balanced-arm sessions from their recorded timers,
 552 showing the three shapes a session takes: the expert beating the stall phrase (seamless handoff), the
 553 expert outliving it (dead air — the tail case above), and the gate staying closed (first token in 0.4 s).
 554 Figure 13 joins the accuracy and latency readouts into the exchange rate actually paid at the med-
 555 ian (speakable subset): +10.6 points of heard accuracy for +0.3 s (never $\rightarrow$ conservative), +6.4 for
 556 +1.0 s (conservative $\rightarrow$ balanced), and +4.1 for +1.1 s (balanced $\rightarrow$ aggressive). The first transition is
 557 both the steepest and the cheapest per second, for two measured reasons: escalation displaces local
 558 decodes that average 4.6 s (and the conservative tier’s escalations are precisely the slowest of them,
 559 hence its P95 shortening above), while the marginal escalations added at higher tiers reach easier
 560 queries whose experts return faster (escalated-session P50 7.5/6.9/5.1 s across the three tiers).

561 8.4 Signal boundary: a third blind spot

562 A fairness challenge to the trap pool — SimpleQA entities are hard to *hear*, conflating knowledge
 563 difficulty with transcription difficulty — motivated auditioning a transcription-fair failure pool: false-
 564 premise questions [Hu et al., 2023], every word common by design ($n = 60$, criterion fixed before
 565 data selection). The audition fails *as a pool* and succeeds as a finding: the small model fails sub-
 566 stantially (.63 text / .47 audio) and the expert would help (.80 adequacy), but the end-of-turn gate
 567 is **blind** to these failures — fire@balanced .18 (vs. trap’s .90), with essentially no score separation
 568 between failed and passed answers (.568 vs. .507). A model answering along a false premise does
 569 not “feel” unable: the mid-layer signal reads *missing knowledge*, not *broken question*. This is the
 570 third documented blind spot, after easy-chat failures and decode-time math: the pre-answer signal is
 571 specifically a *retrieval-failure* detector. FalseQA therefore enters the signal-boundary analysis, not
 572 the tradeoff pools.

573 **A boundary case that data can move: real-time queries.** A fourth query class surfaced in live
 574 demo use: questions whose answer is unknowable at any training cutoff (“what is NVDA trading
 575 at right now”). Unlike false premises, the gate is not blind here — v3 scores sit *on* the balanced

Three live sessions, timestamped (balanced arm; every event time is a recorded live-loop timer)

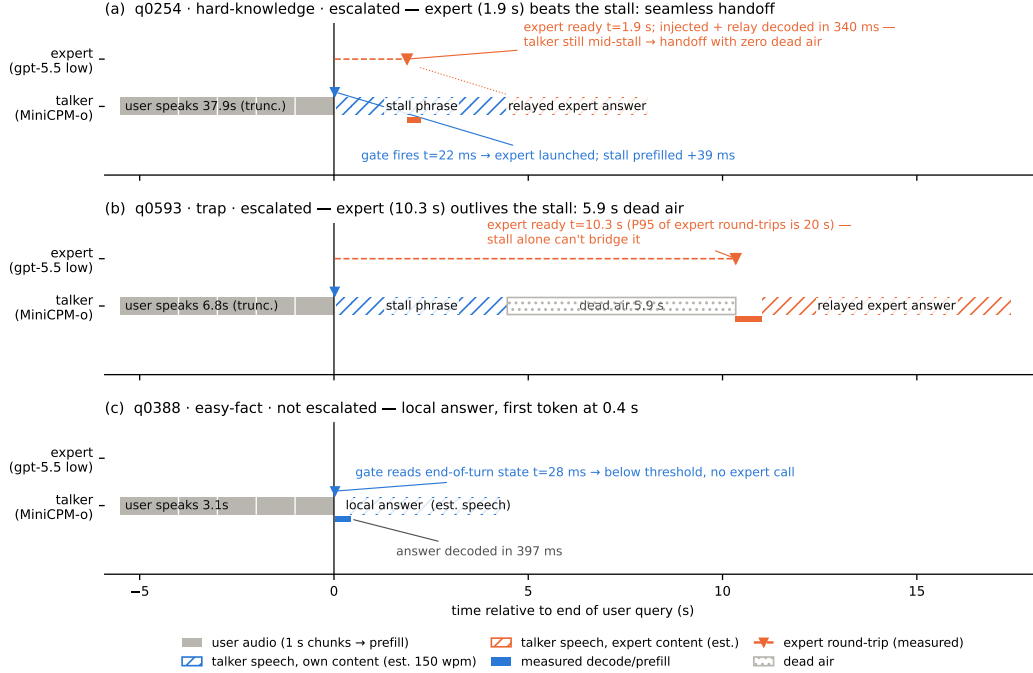


Figure 12: Three real live sessions (balanced arm), replotted from their recorded timers. (a) The expert returns at 1.9 s, while the talker is still voicing the stall phrase: the answer is injected mid-stall and the handoff has zero dead air. (b) The trap-pool case: the expert takes 10.3 s, the stall ends first, and 5.9 s of dead air follow — the P99 signature of Table 9 as a single session. (c) The gate stays closed and the local answer’s first token arrives at 0.4 s. Hatched speech bars are the one estimated quantity (150 wpm; the live loop emits text); every event timestamp is measured.

threshold, flipping with the TTS render (.624 vs. .693 for the same question) — the class was simply absent from every (static-knowledge) training family. Extending the calibration pool with FreshQA [Vu et al., 2023] closes it: fast-changing questions (153, labeled escalate a priori — the current value is unknowable by construction, so a judge against a stale gold would only add noise) plus never-changing in-family controls (150, standard judge labels, fail rate .47). Refit with the frozen v3 feature config, the v4 probe fires on held-out fast-changing questions at .80/.93/1.0 across tiers (v3: .57/.80/1.0) while never-changing held-out rates and frozen-test AUC are unchanged (.877 vs. .879). One calibration subtlety: tier budgets must remain quantiles of the *deployment-mix* scores; quantiling over the augmented train mix (123 extra all-positive rows) silently raised the balanced threshold from .680 to .737, spending the new capability on its own threshold shift. Escalation only pays for this class if the expert can reach live data — the demo’s expert therefore gains a web-search tool, while all measured arms keep the tool-free expert.

9 Transfer Without Training: The Frozen Gate on Six Public Speech Benchmarks

Everything so far is measured on our internal pool, and §4.2 showed exactly why that is not enough: in a mixed pool, most of a router’s apparent margin can be the pool/type shortcut. This section is the out-of-distribution test the audit demands. We take the gate *frozen* — probe weights, feature configuration, and layer choice fit once on the internal calibration mix — and run the complete live streaming loop of §8 on six public speech-QA benchmarks: Speech TriviaQA, Speech Web Questions, Llama Questions, and Reasoning QA from OpenAudioBench [Li et al., 2025] — spoken

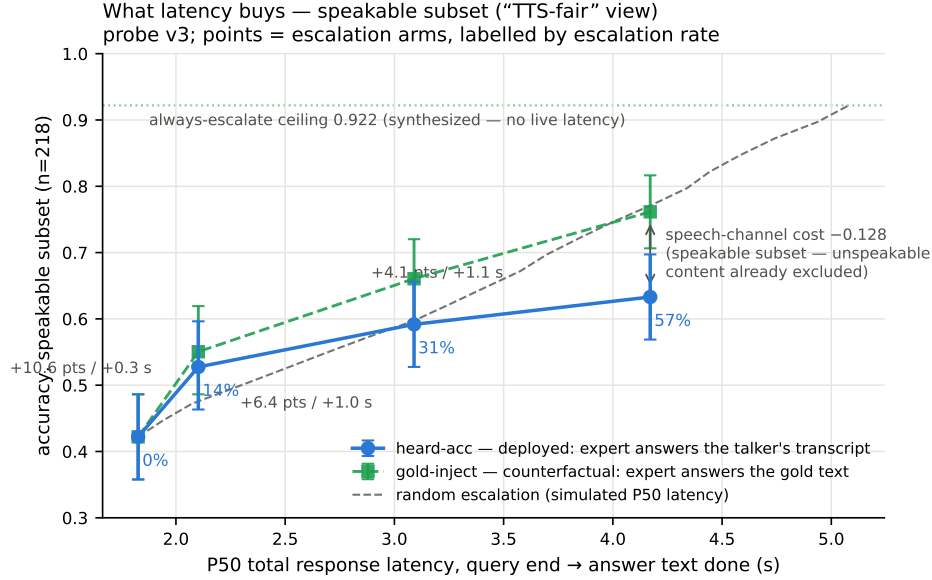


Figure 13: The live accuracy–latency tradeoff on the speakable subset: per-arm P50 response latency against accuracy with 95% paired-bootstrap CIs (Table 8), in the honest heard view (blue) and the channel-controlled gold-inject view (green) of the same sessions — rescoring changes outcomes, not timing, so the two views share their x positions. Segment labels give the median exchange rate on the heard curve; the dotted curve is the simulated-P50 random-escalation reference. The always-escalate ceiling is synthesized from frozen gold answers and has no live latency. Median view — the expert P99 tail is priced in Table 9. Full-pool version: Figure 17.

renderings of TriviaQA [Joshi et al., 2017], WebQuestions [Berant et al., 2013], and the Llama Questions set of Nachmani et al. [2024] — SD-QA (real dialectal human speech, not TTS) [Faisal et al., 2021], and VoiceBench AlpacaEval [Chen et al., 2024, Taori et al., 2023]. All pools use the benchmarks’ official pre-rendered audio — none of it passes through our TTS — and none of their queries ever touches probe training. The deployed probe (v3) is the same L22 end-of-turn read plus two additional zero-latency positions (a rolling 8-token tail and the running user-audio mean), refit on a calibration mix widened to 2,310 queries drawn from public families (SVAMP, HotpotQA, PopQA, MMLU, TruthfulQA-style traps) that are disjoint from — and deduplicated against — the six evaluation benchmarks; the pre-registered guard held with headroom (frozen internal test AUC .860→.879, and feature selection never saw the externals). The only per-pool adaptation is *label-free*: escalation-tier thresholds are quantiles of the probe’s own score distribution on each pool, a normalization the score scale forces (the aggressive threshold is .111 on Llama Questions vs. .603 on SD-QA — a 5× shift for the same probe) and one that consumes no labels, no answers, and no judge calls.

Scoring is each benchmark’s own where a protocol exists: the three OpenAudioBench pools use the official judge verbatim (alias-match grading), AlpacaEval uses VoiceBench’s official 1–5 judge, and SD-QA / Reasoning QA (which ship no replicable protocol) use our reference-anchored judge. Under the official judge our offline chat-mode control *reproduces* the anchors — .712 vs. the official .755 on Speech TriviaQA (250-item subsample + protocol) and .716 vs. .702 on Web Questions — so the official lines in our figures are on-scale, and the live floor’s remaining gap to them is fully accounted (striviaqa: −.048 streaming-loop tax, −.043 subsample/protocol; nothing unexplained).

Table 10: The frozen gate on six public speech benchmarks, live streaming loop, v3 probe, realized escalation rates 15/30/50%. Judge: OAB = the benchmark’s official OpenAudioBench judge; VB = VoiceBench’s official 1–5 judge; ours = reference-anchored (no official protocol exists). Ceiling = GPT-5.5 on gold question text under the same judge; AUC = v3 probe score vs. never-arm local-fail label; official = MiniCPM-o 4.5’s reported number (offline chat mode; our own chat-mode control reproduces it, see text). Per-arm live accuracies carry a replication-noise floor of $\pm .02-.03$ (§9.3).

pool	judge	n	escalation tier				ceiling	AUC	official
			0%	15%	30%	50%			
Speech TriviaQA	OAB	250	.664	.732	.800	.860	.972	.789	.755
Speech Web Questions	OAB	250	.57	.628	.680	.732	.844	.785	.702
Llama Questions	OAB	250	.84	.904	.904	.948	.928	.806	—
SD-QA	ours	200	.51	.600	.720	.795	.930	.792	—
Reasoning QA (zh)	ours	202	.59	.653	.713	.772	.871	.683	—
AlpacaEval (1–5)	VB	199	3.99	3.96	4.23	4.35	4.96	—	4.8

9.1 Accuracy: the mechanism ports, and the anchors fall

Table 10 is the headline: the frozen gate transfers mechanically everywhere — thresholds fire at sane rates, every curve rises monotonically, and the 50%-escalation arm lifts the live floor by +.11 to +.29 across the five QA pools. Three anchor results:

Routing beats scaling the speech backbone. On Speech TriviaQA the 9B duplex model plus the frozen gate at 50% escalation scores **.860 live**, against .755 for MiniCPM-o 4.5’s own official offline number and **.629** for Qwen3-Omni-30B-A3B [Xu et al., 2025b] at $3.3\times$ the parameters (offline chat mode; Kimi-Audio, Kimi Team et al., 2025, scores .419 on the same table). On Web Questions the gated system clears the official number (.732 vs. .702) and lands just under Qwen3-Omni (.749).

On Llama Questions, selective escalation matches the expert at half the cost. The pool is single-type (short English factoids), so no type shortcut exists to exploit, and the local model is strong (floor .84). The probe still splits it cleanly: at 50% the kept-local half scores **.976** vs. **.696** on the escalated half ($z=6.46$) — genuine per-query discrimination — and the expert’s entire advantage concentrates where the probe sends it (escalated half .696 $\rightarrow$.888, McNemar $p<.0001$) while buying nothing measurable on the kept half (.976 vs. .968, $p=1.00$). Selective at 50% reaches .948 vs. .928 for always-escalate (directionally above, $p=.125$ — both near ceiling): escalating everything spends twice the expert calls to buy nothing measurable over the gate.

Router quality must be read as AUC plus matched-rate precision, not raw precision. Precision at a fixed escalation rate is capped at base-fail/rate: on Llama Questions (base fail .16) the cap at 50% is .32, and the v3 gate’s matched-rate precision of .304 is **95% of the theoretical maximum** (recall .93); quoting “.30 precision” without the cap would misread the strongest pool as the weakest. The scale-free readout is transfer AUC against the never-arm fail label: .68–.81 on every external pool (Table 10) vs. .879 on the internal held-out test — the mid-layer signal ports as a generic difficulty signal, with a near-uniform matched-rate precision lift of $\times 1.33-1.35$ across pools rather than pool-specific structure. The cleanest evidence that the signal is difficulty, not surface, is cross-lingual: the calibration expansion is *English-only*, yet it moves Reasoning QA — a *Chinese* execution-reasoning pool — from AUC .621 to **.683** (+.062), the largest per-pool gain of the refit. Per the replication-noise analysis below, the v3 evidence is this offline AUC shift; the corresponding live-arm deltas (+.010–.030) are individually within noise and we do not narrate them.

9.2 Latency: the probe escalates the slowest decodes first

The latency curves of Figure 14 have a shape worth explaining honestly, because at first sight it looks like a measurement error: on the easy pools the per-arm P50 is *non-monotonic* — on Llama Questions it *falls* from 1.52 s (never) to 1.17 s at 15% escalation before rising again (1.60/1.42 s at 30/50%). This left-fold is a verified median-of-a-mixture mechanism, not noise: the probe prefer-

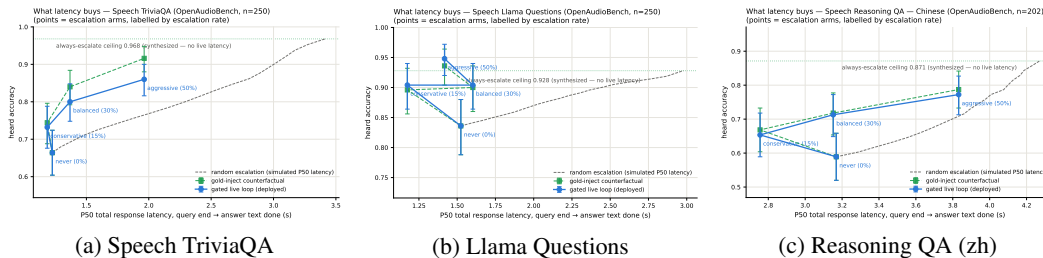


Figure 14: Accuracy vs. per-arm P50 response latency on three external pools (frozen gate, live arms; official judge on a–b, ours on c). The random reference simulates a random mixture at each rate from per-id *measured* latencies (400 sims $\times$ 21 rates). On (a) and (b) the gated curve dominates random in *both* axes: on Speech TriviaQA random needs ~ 3.4 s at the median to reach the accuracy region the gate reaches at 2.0 s, and on Llama Questions random at 50% sits at ~ 2.2 s/.88 vs. the gate’s 1.42 s/.948. On (c) escalation truncates the latency *tail* (P90 13.25 $\rightarrow$ 10.94 s). Remaining family figures: Appendix B.

ententially escalates exactly the queries whose *local* decode is longest. On Speech TriviaQA the 38 queries escalated at 15% had a never-arm local P50 of 1.711 s vs. 1.198 s for the 212 kept — and, measured in the never arm, local accuracy .32 vs. .73: the small model’s signature failure mode on retrieval pools is hedging at length, so wrong answers are slow answers (decode time and answer length are the same variable, $r=-.97$), and a wrongness-selecting probe strips the slow tail of the local mixture as a side effect. Controls: the probe does not key on length directly ($\text{corr}(\text{score}, \text{length}) = +.05$; conditioned on wrongness the score–length coefficient is ≈ 0), and on SD-QA — where wrong answers happen to be short and confident — the same probe selects wrongness identically and produces *no* fold. Figure 15 decomposes the effect: the fold lives only in the median — the *mean* is monotonic (1.65/1.81/1.96/2.06 s vs. P50 1.52/1.17/1.60/1.42 s) — and the kept-local P50 falls 1.52 $\rightarrow$ 0.96 $\rightarrow$ 0.86 $\rightarrow$ 0.43 s as the probe strips slow decodes while escalated rows pay a flat ~ 3 s expert round-trip. We report the mean alongside the median for exactly this reason; the honest statement is that on these pools escalation buys accuracy *and*, at the median, speed. The individual dip depth is within noise (striviaqa -41 ms, bootstrap CI $[-285, +254]$ ms) but its sign is deterministic — identical probe scores select the identical escalated set on every rerun.

On Reasoning QA the shape inverts usefully: P50 rises gently but the *tail improves* — P90 drops from 13.25 s to 10.94 s under escalation. The mechanism is that the local model’s chain-of-thought is *spoken*, so every reasoning token enters the response clock (escalated-set local decodes P50 4.38 s, tail 13 s+), while the expert’s chain is hidden: its median completion is 144 tokens of which only 57 characters return, behind a flat ~ 3.7 s round-trip. On reasoning pools the gate replaces the longest spoken chains with short round-trips — escalation truncates the latency tail rather than fattening it.

9.3 Where transfer fails, and what the live curves can and cannot show

AlpacaEval is a negative result, mechanism measured. On VoiceBench AlpacaEval the gate does *not* beat random: aggressive 4.35 vs. random-at-50% ≈ 4.45 . The failure is selection, not escalation — queries the gate selected score 3.90 on the never arm vs. 3.98 for those it skipped (no discrimination), while escalated rows still gain (3.90 $\rightarrow$ 4.71) purely because GPT-5.5 writes better long-form answers; random picks would harvest the same gain. Open-ended instruction following has no “the model lacks this fact” event for a retrieval-failure detector to read, and the score range is compressed (1.0 point vs. 40 accuracy points on our pool). Two scale caveats make the absolute level honest: the official 4.8 is an *offline chat-mode* number — our chat-mode control reproduces it at 4.86 — and the entire chat-vs-live gap (4.86 vs. 3.99) is the duplex loop’s spoken-reply style plus a 512-token cap, so only the live arms are comparable to each other.

Web Questions and SD-QA are flat under the v3 refit, offline (AUC $+0.006/+0.017$) and live alike. On Web Questions a specific share of the blame is label fidelity rather than the model: its strict Freebase-alias grading decouples “wrong” from “uncertain” — the wrong-answers-are-longer sig-

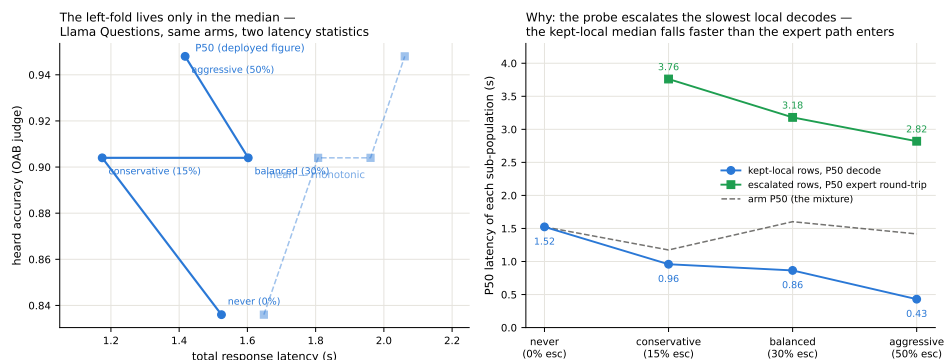


Figure 15: Decomposition of the Llama Questions latency fold. Left: the fold exists only in the median — the mean is monotonic. Right: kept-local P50 falls 1.52→0.43 s across tiers as the probe strips the slowest local decodes, while escalated rows pay a flat ~3 s expert round-trip; the arm median folds left and then rises as the expert mass reaches it.

686 nature is null there for every model we measured, including GPT-5.5 on gold text — so a portion
 687 of its labels is protocol-wrong that no state-reading probe can predict, and even the gold-text expert
 688 ceiling is .844.

689 **The deployed view understates selection on easy pools.** The external channel cost is small — on
 690 the v3 traces the aggressive-arm heard-vs-gold-inject gap is .056/.036/.035 on striviaqa/swebq/sdqa
 691 vs. .175 on the internal pool (.128 on its speakable subset, §8.3), exactly as predicted for short open
 692 questions with no option walls — but on easy homogeneous pools even a small tax eats a thin per-
 693 query margin, and part of the deployed-view “gate ≈ random” on Speech TriviaQA is channel cost:
 694 in the channel-controlled gold view the gate clears its paired random line by +.100 at the aggressive
 695 arm (.916 vs. .816). SD-QA meanwhile carries the one property no TTS pool has — real dialectal
 696 human speech — and the escalated subset still converts at .82–.84 with expert P50 3.1–3.4 s: the
 697 payload survives real speech.

698 **Replication-noise discipline.** Re-running the same query with the same audio through the live loop
 699 flips the judge verdict 2.3–18.8% of the time depending on pool (Llama Questions .023, SD-QA
 700 .188), implying a paired SE of .009–.028 on any arm-vs-arm accuracy delta. Against that floor,
 701 all 18 per-arm live v2→v3 deltas are individually non-significant (the largest, striviaqa balanced
 702 +.036, gives McNemar $p=.16$). We therefore treat sub-3-point live deltas as unresolvable in a single
 703 sweep and rest every probe-improvement claim in this section on the variance-reduced instruments
 704 — offline AUC over the full score distribution and matched-rate precision — with the live curves
 705 serving as the end-to-end existence proof, not the measurement.

706 10 Discussion

707 10.1 Design consequences: the two-stage gate

708 The evidence converges on a two-stage design:

- 709 • **Stage 1 (prefill, effectively free):** a mid-layer (~60% depth) last-token probe — 20/45 ms,
 710 before the first token, and the only signal that survives duplex fine-tuning *and* both input
 711 modalities (§6.5). Open engineering item: score-scale calibration for threshold (rate) trans-
 712 fer.
- 713 • **Stage 2 (optional, post-draft):** $p(\text{True})$ -post on the drafted answer — the strongest single
 714 signal (0.899); the draft doubles as the distilled escalation query and as the fallback an-
 715 swer. On audio input, never ask over raw audio context: run $p(\text{True})$ on the model’s own
 716 transcript (“repeat-then-judge,” §6.3).

10.2 Positioning vs. query-side routing

Query-side routers [Ong et al., 2024, Ding et al., 2024, Chen et al., 2023] decide before generation from query features, and the big model then answers the user directly. Neither transfers to full-duplex voice: the user talks to one voice whose session state lives in the small model, and query-side features are precisely the “type shortcut” our audit quantifies (pool oracle 0.715 vs. probe 0.822 in-distribution) and LOPO breaks. The signal must come from the answering model itself — consistent with Li et al. [2026]’s attribution of router failures to model-recall gaps. A head-to-head comparison of query-feature routers vs. internal-signal gates under distribution shift, on their standardized protocol, is natural future work.

10.3 Result injection: resolved by the same signal

Step 2 — returning the thinker’s result into the live duplex session — initially looked like a research problem of its own: the streaming smoke test showed the talker pushing back on conflicting injected results. The quantified answer (§8.2) is that for *gated* escalations the problem largely dissolves, and for a principled reason: the same low-confidence state that fires the gate predicts relay compliance (comply .42 → .75 across gate-score quartiles; 1.00 on the deployed wrong-model × correct-expert workload). The gate and the relay are two uses of one signal. Where the state is confident, framing — not ventriloquism — is the mechanism (authority framing .53–.79 vs. seeding’s backfire), and the residual ceiling is the swallow rate: an unconfident talker relays a *wrong* expert at .64 with zero pushback, so cascade accuracy is bounded by expert quality.

10.4 Three failure species, one perceivability axis

The accumulated blind-spot results organize into a taxonomy of *why* a model fails, ordered by whether the failure is an observable event in the model’s own processing. **Retrieval-type** failures (“I don’t know this entity”) produce an observable empty lookup at prefill — pre-answer signals work (trap fire .90). **Execution-type** failures (“stuck mid-solution”) only come into existence during decoding — decode-time signals see them (math entropy@16 .860) and pre-answer signals cannot. **Metacognitive-type** failures (“doesn’t know the question itself is broken”) never occur as an event anywhere in the model’s experience — like a schoolkid confidently “verifying” Goldbach’s conjecture, the model answering a false-premise question executes smoothly along the wrong frame, and every internal signal is blind (FalseQA fire .18, no score separation; §8.4). This mirrors the human feeling-of-knowing literature, which finds metacognition accurate for memory retrieval and unreliable for insight problems. The taxonomy explains the two-stage gate architecture (§10.1 rationale: prefill probe for species 1, decode monitor for species 2), scopes the claim — the pre-answer signal is a retrieval-failure detector, not a general oracle — and notes that query-surface routers miss species 3 as well.

10.5 Demonstration system

The measured pipeline runs as an interactive web demo: live microphone audio streams to the duplex model on a single H100, the end-of-turn probe read costs ~54 ms against the balanced threshold, and escalated turns stall, consult the expert, and *speak* the relayed answer through the model’s own TTS head. Two demo-only extensions go beyond the measured arms: the expert gains a web-search tool — real-time queries such as stock prices, which the v4 FreshQA refit (§8.4) fires on at .80–1.0 across tiers, then return live values no cutoff-bound model can produce — and the loop accepts *backchannel-aware barge-in*: sustained speech over the talker ducks playback, a ~1 s ASR pass classifies the burst, and backchannels (“ok”, “mm-hm”; an English+Chinese lexicon) return the floor with zero loss while anything else aborts generation and seeds the next turn. All measured arms in this paper keep the tool-free expert and the turn-based loop; the demo will be made available upon publication.

10.6 Limitations

(i) $p(\text{True})$ prompt sensitivity: one phrasing each (pre/post). (ii) The audio arm is validated on real dialectal speech (SD-QA), but scripted read speech and one dialect split; spontaneous conversational speech (disfluencies, self-repair) remains uncovered. (iii) Input-side scope: formula-symbolic content (spoken LaTeX) is destroyed by *any* TTS render before the system hears it, so the audio arm is scoped to speakable content via pre-registered input-side filters (§8.3); this limit is distinct from the transcription-uplink loss, of which a stronger cloud ASR diagnostically recovers 38%. (iv) The introspection-collapse finding is MiniCPM-scoped (two generations of one family); the pre-registered second-family test now passes on a hybrid-Mamba architecture for mid-band readability and frozen-recipe transfer (§5.4), but the full matched-pair attribution against its raw backbone and the live-loop port were not repeated there. (v) The frozen test split is reused across signal comparisons (signals were pre-specified; a fresh test set would be cleaner). (vi) Labels come from a single judge family, bounded by a stronger re-judge (deltas ≤ 0.010) but without human labels. (vii) The live sweep (§8) is one session per query per tier, with a documented replication-noise floor of ± 0.02 – 0.03 per arm; no claim rests on sub-3-point live deltas. The measured sweep ran with text-mode talker output (`generate_audio=False`); the relay is validated at the utterance level and latencies are text-complete — a conservative accounting, since in the spoken system the stall filler plays the moment the gate fires; the demo system (§10.5) speaks through the native TTS head without touching the probe context or the frozen thresholds. The native duplex listen/speak head remains unused throughout: the probe is calibrated on simplex streaming states, and its validity under the true duplex loop is an open experiment.

Future work. In rough order of leverage: (a) an audio-direct-to-expert live arm — the gold-inject view prices the transcription channel at $-.175$ on the internal pool, an order above the noise floor, and it is the one lever a single live sweep can resolve; (b) true duplex integration — driving the native listen/speak head and re-validating the L22 read under the duplex loop; (c) the Nemotron live port and its trained-checkpoint ablation, completing the second-family replication; (d) a *trained* gate head as an upper-bound contrast for the zero-training claim; (e) the remaining ten SD-QA dialect splits as a cross-dialect robustness table; (f) protocol-level comparison against preference-trained routers on LLMRouterBench [Li et al., 2026]; (g) a $p(\text{True})$ prompt-sensitivity sweep; and (h) an injection-payload study (does “answer + one-sentence justification” restore math relay compliance where bare numbers fail).

11 Conclusion

Zero-training self-knowledge in a small full-duplex model exists, is strong, and is modality- and layer-fragile in systematic, duplex-fine-tune-induced ways. The standard readouts — final-layer probes and verbalized self-evaluation — each have one blind modality, and both blind spots are created by the duplex fine-tune; the mid-layer representation underneath survives everything we tested. Read there, the resulting gate decides before the first output token, transfers from cheap text calibration to speech on end-to-end models, and as a system dominates the model’s own thinking mode on both accuracy and latency. And when the thinker’s answer comes back, the same signal that fired the escalation predicts that the talker will actually say it: run live end-to-end, the gated duplex system climbs monotonically from its honest streaming floor ($.422 \rightarrow .633$ at 57% escalation on the speakable subset), with the remaining gap to the offline curve itemized into named, measured taxes rather than left as mystery — and the frozen gate transfers unchanged to six public speech benchmarks, where at 50% escalation it lifts the 9B system past models $3\times$ its size. Gate and relay are one mechanism; what remains is engineering the channel around it.

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982 A Full-pool live figures

983 Figures 16 and 17 repeat the headline live figures of §8.3 on the *full* frozen test pool ($n=240$), i.e.
 984 without the pre-registered speakable-content filter; numbers for both views appear in the lower block
 985 of Table 8. Figure 18 is the replication-noise audit behind the $\pm.02$ – $.03$ floor quoted throughout.

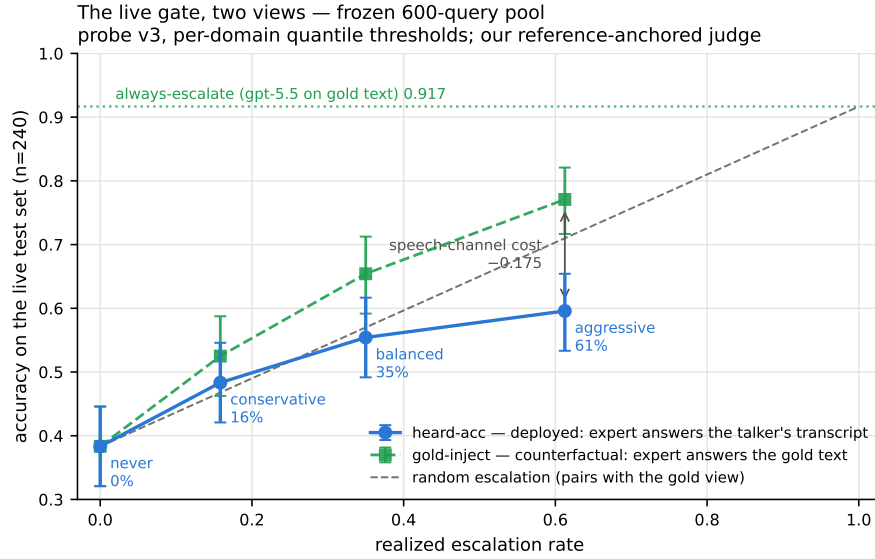


Figure 16: Dual-view live tradeoff, full pool ($n=240$); companion to Figure 11.

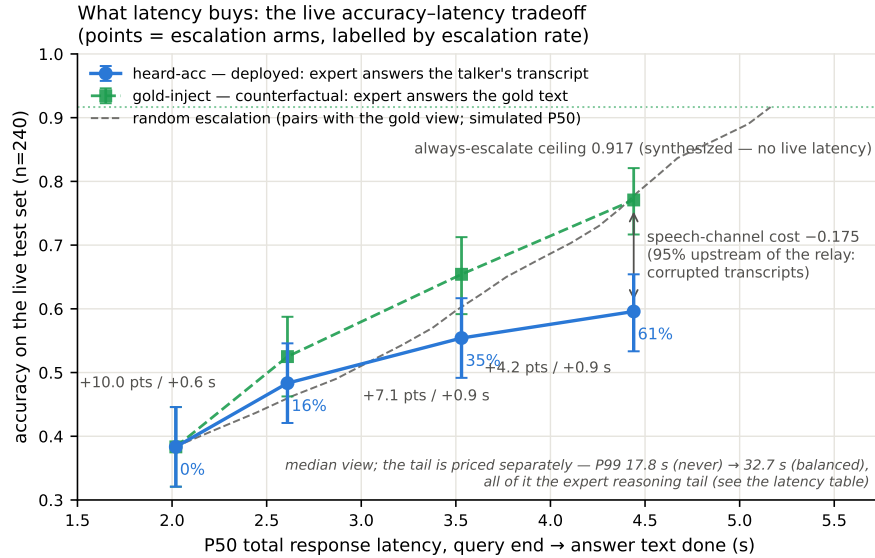


Figure 17: Accuracy–latency tradeoff, full pool; companion to Figure 13.

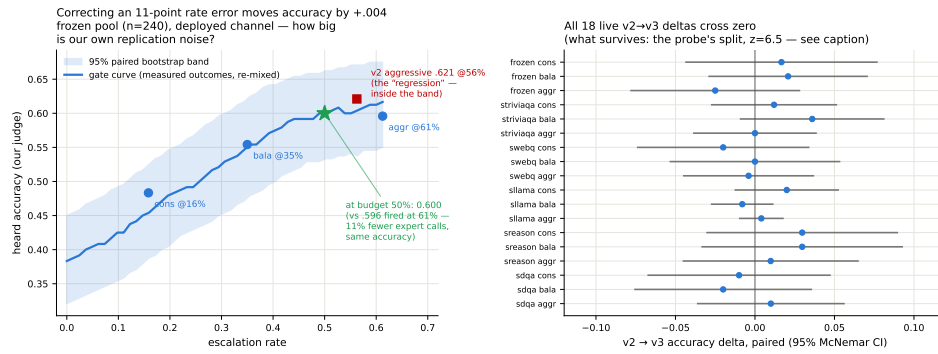


Figure 18: Replication-noise audit: same query, same audio, re-run through the live loop across arms. Kept-local judge verdicts flip 2.3–18.8% of the time depending on pool; repeated escalation flips 0.7–10.6%; implied paired SE on an arm-vs-arm delta .009–.028. This floor is why the paper interprets no sub-3-point live delta.

986 B External-benchmark figure set

987 Dual-view and accuracy–latency figures for all six external families (frozen v3 gate, live streaming
988 loop; §9).

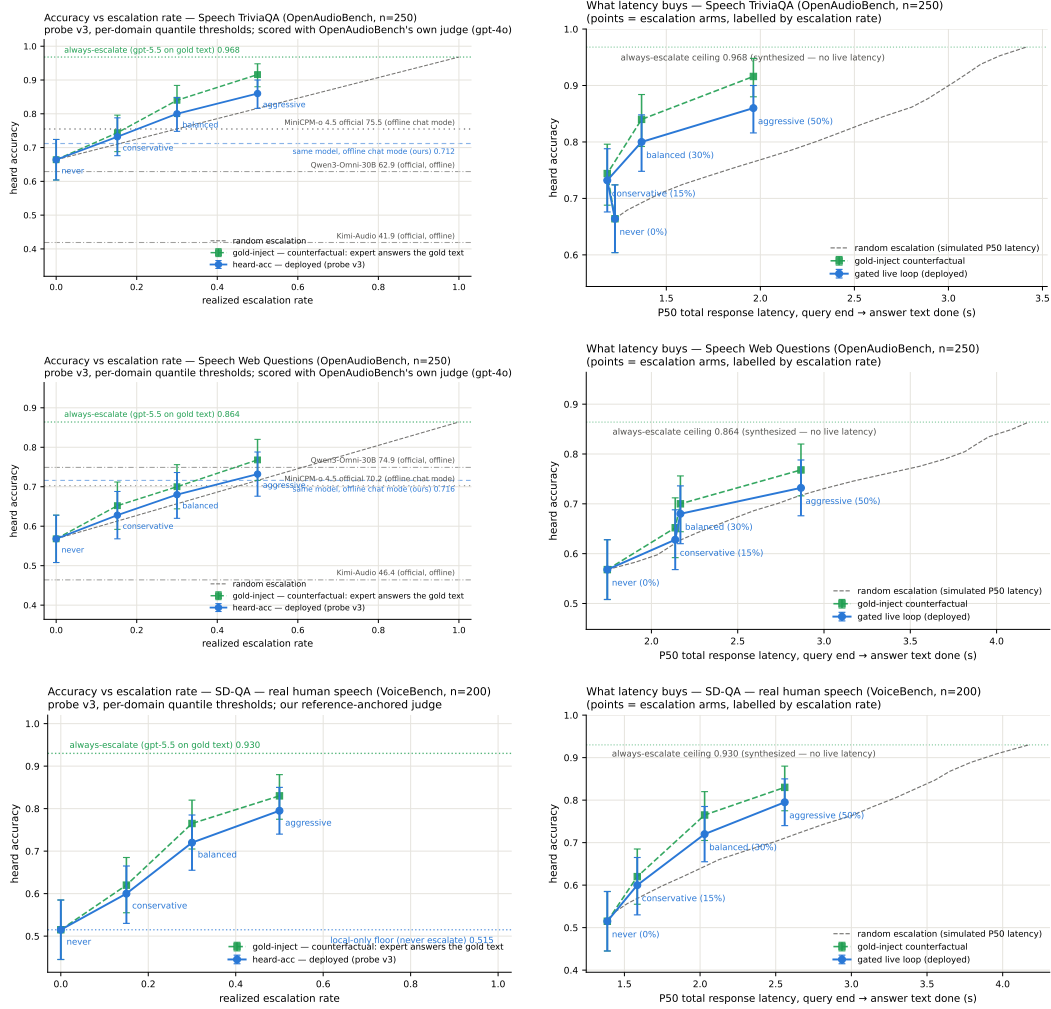


Figure 19: Speech TriviaQA, Speech Web Questions, SD-QA: dual-view (left) and accuracy–latency (right).

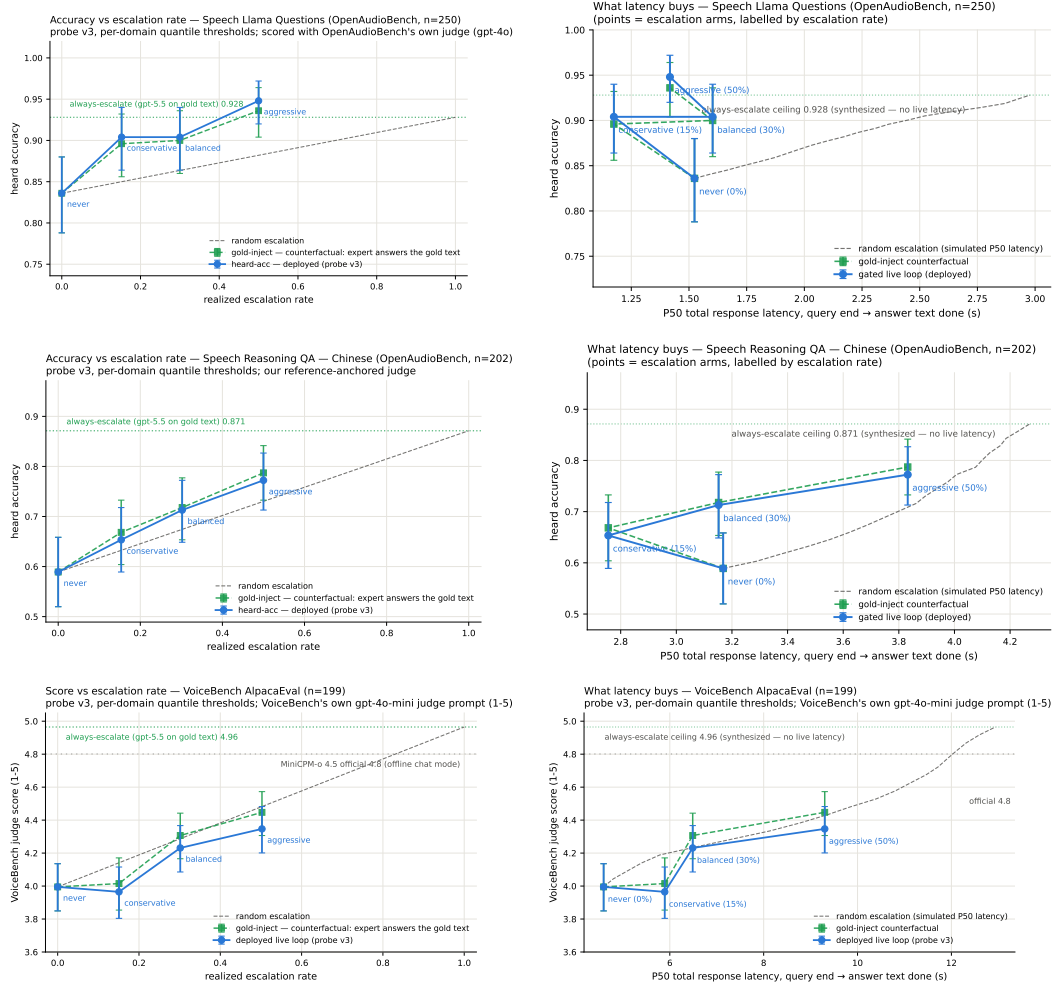


Figure 20: Llama Questions, Reasoning QA (zh), VoiceBench AlpacaEval: dual-view (left) and accuracy-latency (right). AlpacaEval is the measured negative result of §9.3.

989 C Judge and self-evaluation prompts

990 All adequacy labels use the following fixed system prompt (gpt-5.4-mini, low reasoning effort,
991 OpenAI Structured Outputs with a strict {reason, adequate} JSON schema; the judge never sees
992 which system produced the answer):

```
993     You are a strict, fair grader of a small language model's answers.  
994     You are given a QUESTION, an optional REFERENCE answer, and a  
995     CANDIDATE answer produced by a small model. Decide whether the  
996     CANDIDATE is ADEQUATE, meaning ALL of: (1) factually correct;  
997     (2) any reasoning shown is valid; (3) it actually and completely  
998     answers the question that was asked. For multiple-choice questions  
999     the candidate must land on the correct option (by letter or by  
1000    unambiguously stating the option's content). A partially correct,  
1001    hedged, refused, off-topic, or truncated answer is NOT adequate. If  
1002    a REFERENCE is provided, grade against it. Be strict but do not  
1003    penalize harmless differences in format or extra correct detail.
```

1004 The verbalized self-evaluation baselines (§4) use one fixed phrasing each; $P(\text{Yes})$ is read from the
1005 first generated token's distribution:

```
1006     [pre] I am going to show you a question. Do NOT answer it. Judge  
1007     honestly whether you yourself would answer it correctly. Question:  
1008     {q} Would you answer this question correctly? Reply with exactly one  
1009     word: Yes or No.  
1010     [post] Here is a question and a proposed answer. Question: {q}  
1011     Proposed answer: {a} Is the proposed answer correct? Reply with  
1012     exactly one word: Yes or No.
```

1013 D Reproducibility

1014 All sampling and splits use a fixed seed (42); the calibration/test split is frozen once and the test split
1015 is touched only by the evaluation scripts. Environment pins are listed in §3 (torch 2.8.0; transformers
1016 4.51.0 for MiniCPM-o 4.5, with per-model pins for the comparison backbones); all experiments
1017 run on single-H100 instances. Probe artifacts (weights, feature configuration, per-pool label-free
1018 thresholds), the judge prompt above, all trace parquets, and the per-experiment execution log —
1019 including per-run GPU and API spend, on the order of \$500 total across the project — ship in the
1020 project repository, released upon publication.

1021 **TODO (process only — delete before submission)**

- 1022 • **[TODO: P1 — compress main text to 8 pages (currently far over); section-order and**
- 1023 **cut decisions at the 2026-08-26 meeting. Cut priority per RTCA positioning: generic**
- 1024 **uncertainty exposition first; live / transfer / duplex evidence last]**
- 1025 • **[TODO: P2 — incorporate peer + meeting feedback]**
- 1026 • **[TODO: P3 — submission day (8-28): delete this section; final anonymization grep**
- 1027 **(names, rhe9527, modal URLs, gmail); decide public-repo → private; OpenReview**
- 1028 **form + upload (deadline 8-29 AoE, moderation hard limit Fri 8-28)]**